ICMA CENTRE

IC103

Introductory Economics for Business and Finance

Lecture 3: Theory of Production and Costs

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Reading

Sloman, J., Wride, A. and Garratt,D (2018) “Economics” (10th edition).

https://ebookcentral.proquest.com/lib/reading/reader.action?docID=6824443#

- Section 6.1 except Boxes 6.1-6.3 (Box 6.4 optional)
- Section 6.2except Box 6.5 (Box 6.6 and “Looking at Maths” optional)
- Section 6.3 up to bottom of page 165 (i.e., stop at Box 6.8)
- Section 6.4 only pp. 172-173
- Section 6.5 pp. 176-180 (except “Looking at Maths on pp. 179)
- Section 6.6 up to left part of page 182

Section 1: Background to Supply

1.1 Theory of Production: Intro

Theory of Production: How Firms Use Inputs to Produce Output

The main objective of firm (Theory of the firm): to maximize profit

$$Profit\ (T\Pi) = Total\ Revenue\ (TR) - Total\ Cost\ (TC)$$

Inputs: Fixed vs Variable factors of production

Fixed factors of production

- Inputs that **cannot be changed** in the short run.
- Examples: buildings, machinery, equipment.



Buildings



Machinery



Equipment

Variable factors of production

- Inputs that **can be changed** in the short run.
- Examples: labour, raw materials, electricity.



Labour



Raw materials



Electricity



Other inputs (e.g. water)

Whether an input is fixed or variable depends on the time period considered.

Theory of Production: Short Run vs Long Run

Short run

- At least **one input is fixed**.
- The firm can change some inputs, but not all.



Long run

- **All inputs** can be **changed**.
- The firm has enough time to adjust its production capacity.



Example: A bakery can hire more workers quickly, but expanding the building takes much longer.

The length of the short run is different for different firms and industries.

Theory of Production: Diminishing Marginal Returns

- **The law of diminishing marginal returns**, also called diminishing marginal productivity or simply diminishing returns, states that…

when more units of a **variable input** are added to **fixed inputs**, there comes a point when each additional unit of the variable input produces less extra output than the previous one.

Example: workers in a small bakery

Fixed inputs: bakery space, ovens, equipment
Variable input: number of workers

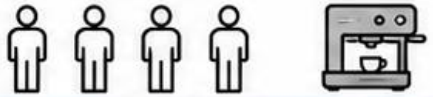
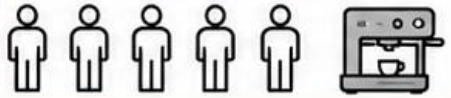
Number of workers	1	2	3	4	5
Total output (loaves per hour)	20	50	75	90	98
Extra output from additional worker	+20	+30	+25	+15	+8



extra output (marginal product) starts to fall

Diminishing Marginal Returns: Coffee-Shop Example

Example: too many baristas, one coffee machine

Number of baristas	Baristas and coffee machine (fixed input)	Total output (coffees per hour)	Extra output (from additional barista)
1		20	–
2		45	+25
3		62	+17
4		70	+8
5		73	+3

Extra output falls as more baristas are added.

What this shows

- One coffee machine is the fixed input.
- Baristas are the variable input.
- The fixed equipment becomes a bottleneck, so each extra worker adds less output.

Diminishing marginal *returns*: total output may still rise, but extra output falls.

Mentimeter EM

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Add Menti slides

A coffee shop has one coffee machine. As more baristas are added, what does diminishing marginal returns mean?



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0	0	0	0
Total output immediately falls	Each additional barista eventually adds less extra output	The coffee machine becomes a variable input	Every additional worker adds the same amount of output

Waiting for participants

Show correct answer

Theory of Production: The Production Function

- **The production function** shows the mathematical relationship between the quantity of output produced and the inputs used in production.

$$Q = f(X_1, X_2, \ldots, X_n)$$

where:

- Q is the quantity of output produced.
- X_i is the ith input used in production.
- Examples of inputs: labour, capital, raw materials and land.

Example: a coffee shop with one espresso machine

Fixed input: one espresso machine and counter space.

Variable input: number of baristas.



Fixed input (one espresso machine and counter space)

Variable input (number of baristas)

Output (coffees produced per hour)

Number of baristas	1	2	3	4	5
Total output (coffees per hour)	20	45	62	70	73
Extra output from additional barista	+20	+25	+17	+8	+3

Output depends on the quantity and combination of inputs used and how productively they are combined.

Theory of Production: Short Run and Long Run

For simplicity, we now focus on two inputs: labour (L) and capital (K).

$$Q = f(L, K)$$

where Q = output, L = labour, K = capital

Short run

- Capital (K) is fixed.
- Labour (L) can vary.
- As labour increases, total output initially rises.
- Eventually, marginal product falls because of diminishing marginal returns.



Long run

- Both labour (L) and capital (K) can vary.
- The firm can adjust all inputs.
- The firm can change its scale of production.
- This leads to economies and diseconomies of scale.



Note: capital K means the physical equipment and facilities used in production.

Section 1: Background to Supply (Short-run)

1.2 The Short-run Theory of Production

Short-run production: Total, Average and Marginal Product

Three key measures of short-run production

- **Total Product (TP)** – total output produced.
- **Average Product (AP)** – output per unit of the variable input.
- **Marginal Product (MP)** – extra output from one additional unit of the variable input.

A simple way to remember them is:

- Total Product means total output.
- Average Product means output per worker.
- Marginal Product means the extra output from one more worker.

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Which measure tells us the extra output produced by one additional worker?

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0	0	0	0
Total Product	Average Product	Marginal Product	Total Cost

Waiting for participants

Show correct answer

Short-run production: definitions and formulae for TP, AP and MP

- **Total Physical Product (TPP)** — also called Total Product (TP)

Total quantity of output produced from a given amount of inputs.

- **Average Physical Product (APP)** — also called Average Product (AP)

Output produced per unit of the variable input.

$$APP = \frac{TPP}{Q_V}$$

where Q_V is the quantity of the variable input used.

- **Marginal Physical Product (MPP)** — also called Marginal Product (MP)

Extra output produced by one additional unit of the variable input.

$$MPP = \frac{\Delta TPP}{\Delta Q_V}$$

- In our farm example, labour is the variable input, so $Q_V = L$.

Farm Production Example: Calculating TPP, APP and MPP

$$Q = f(L, K)$$

L: number of workers (labour input)
K: capital (fixed input)

	Number of Workers (*Lb*)	*TPP*	*APP* (=*TPP*/*Lb*)	*MPP* (=Δ*TPP*/Δ*Lb*)
(a)	0	0	–	
				3
	1	3	3	
				7
	2	10	5	
(b)				14
	3	24	8	
				12
(c)	4	36	9	
				4
	5	40	8	
				2
	6	42	7	
(d)				0
	7	42	6	
				–2
	8	40	5	

= 10 - 3

= 36 - 24

= 40 - 42

How to read the table
TPP = total output
APP = output per worker
MPP = extra output from one additional worker

Diminishing marginal returns begin when MPP starts to fall.

Diminishing marginal returns begin when the fourth worker is added, because MPP falls from 14 to 12.

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A farm produces 24 tonnes with 3 workers and 36 tonnes with 4 workers. What is the marginal product of the 4th worker?

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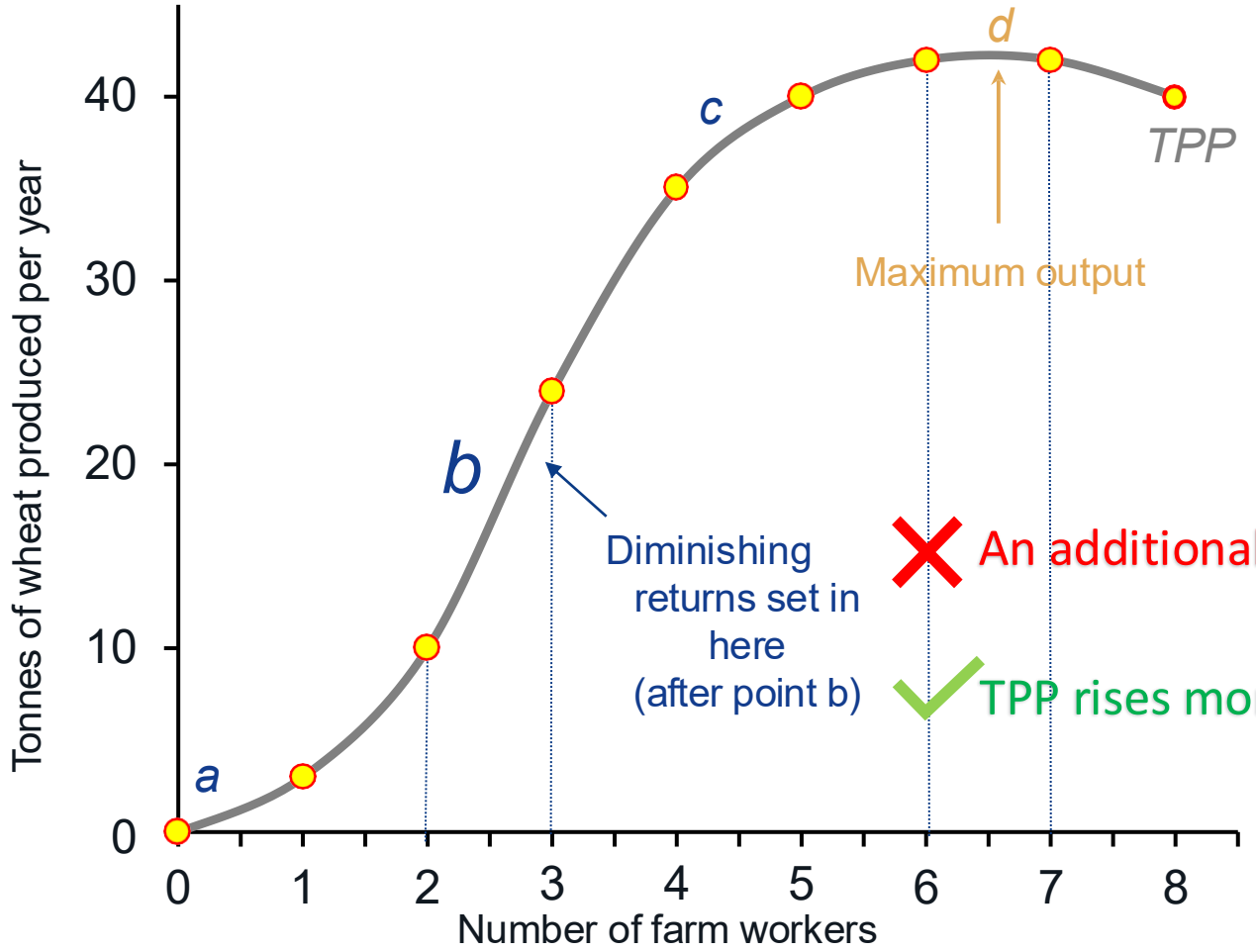
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0	0	0	0
9 tonnes	10 tonnes	12 tonnes	36 tonnes

Waiting for participants

→ Show correct answer

Farm Production: The Total Product Curve



Number of workers	TPP
0	0
1	3
2	10
3	24
4	36
5	40
6	42
7	42
8	40

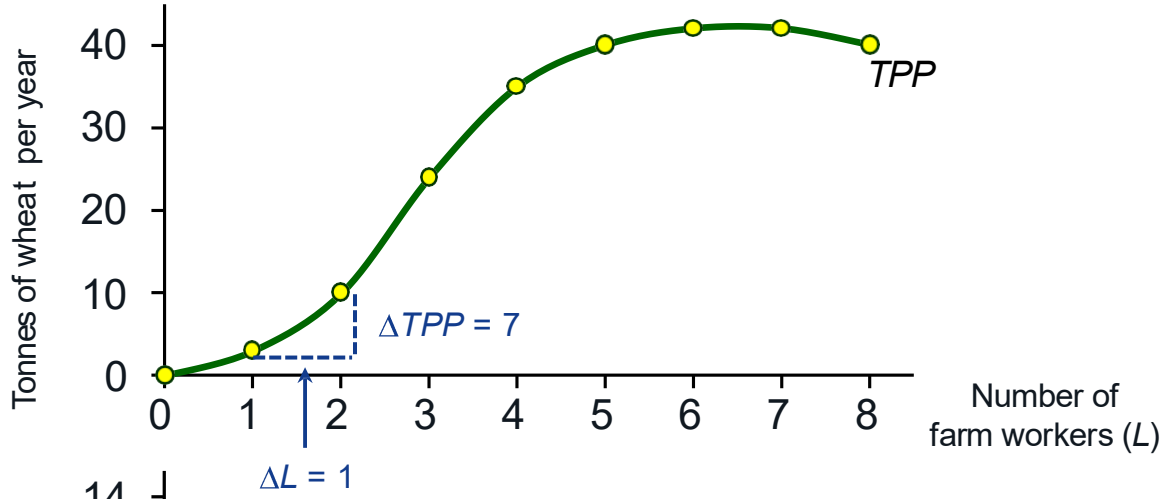
✗ An additional worker adds no extra output

✓ TPP rises more slowly and may eventually fall

Remember: diminishing marginal returns begin before total output starts falling

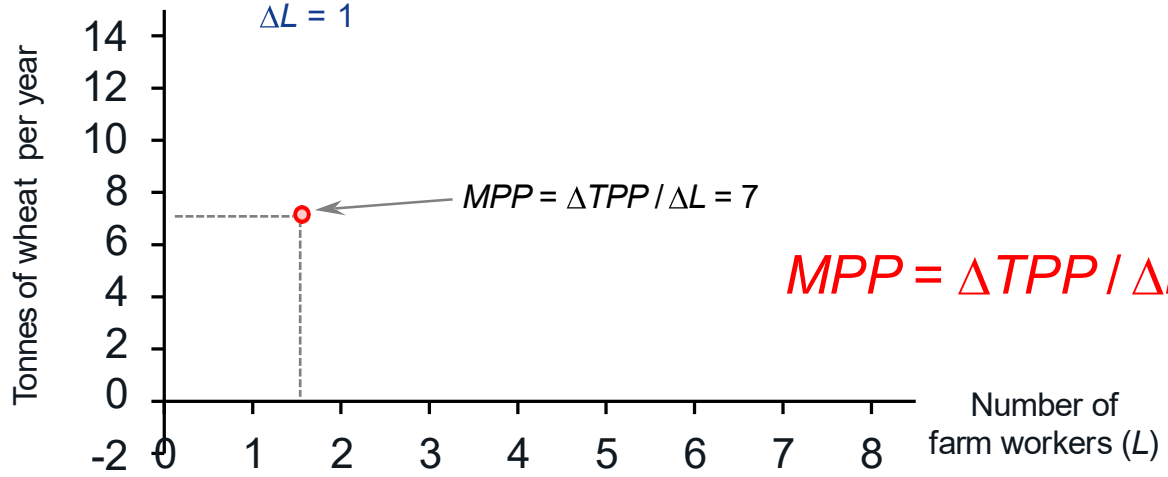
Farm Production: Marginal Product and the Slope of TPP

Total Physical Product (*TPP*):



Marginal Physical Product (*MPP*):

how much total output changes when labour changes

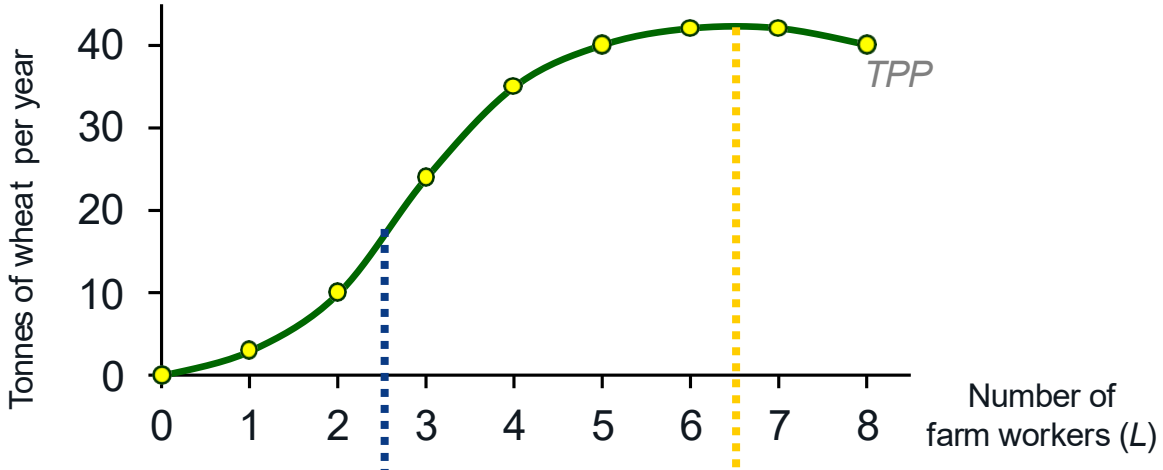


When the TPP curve is steep, marginal product is high. When the TPP curve becomes flatter, marginal product is falling.
Marginal product and the shape of the total product curve are closely connected.

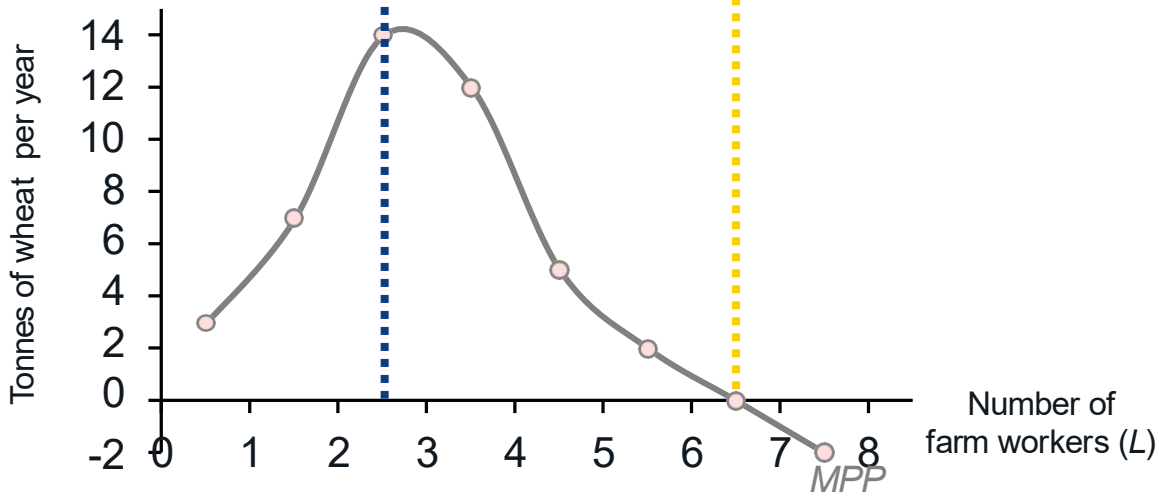
MPP = ΔTPP / ΔL--- the slope of TPP

Farm Production: The TPP and MPP Curves

Total Physical Product (*TPP*)



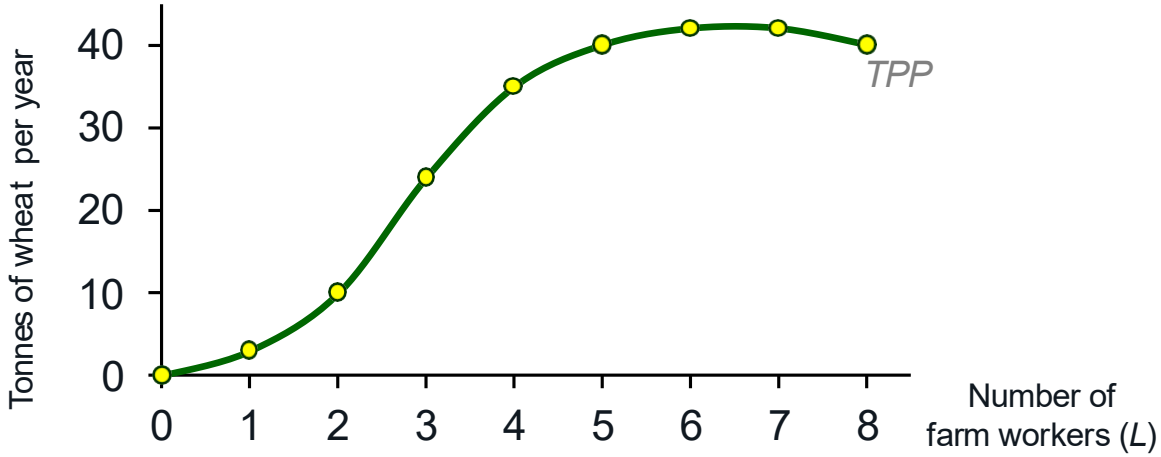
Marginal Physical Product (*MPP*)



As long as MPP remains positive, total product continues to rise.
When MPP reaches zero, total product reaches its maximum.
If MPP becomes negative, total product starts to fall.

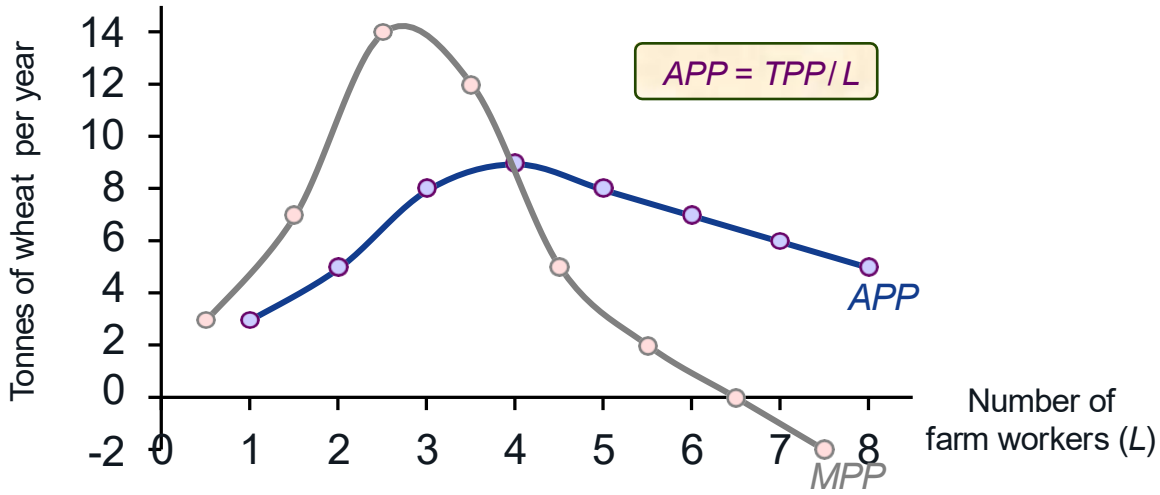
Farm Production: Adding the APP Curve

Total Physical Product (*TPP*)



Marginal Physical Product (*MPP*)

Average Physical Product (*APP*)
average output per worker



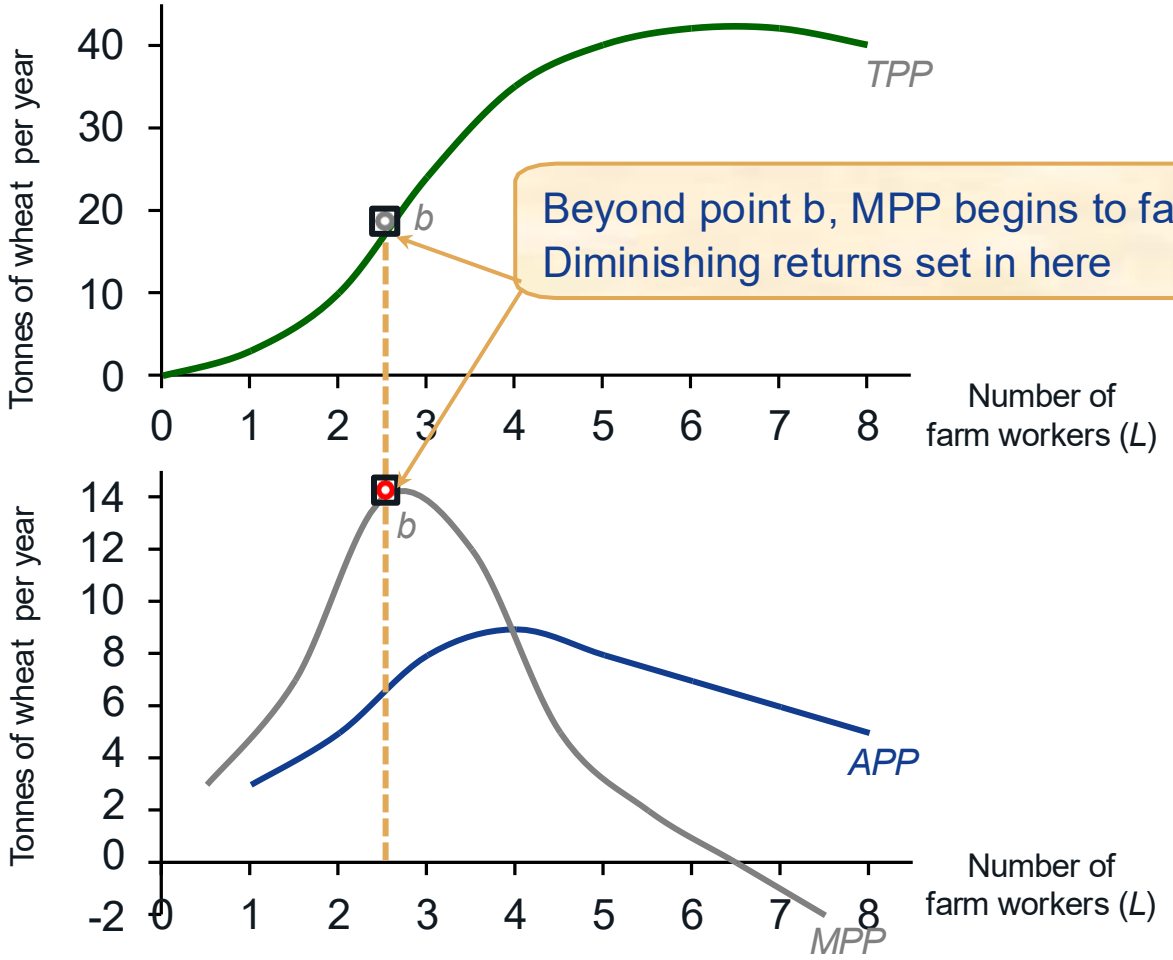
When marginal product is above average product, average product rises. When marginal product is below average product, average product falls.

Tip to understand:
If your next mark is above your current average, your average rises. If your next mark is below your current average, your average falls.

Farm Production: Diminishing Marginal Returns

MPP = ΔTPP / ΔL;
the slope of TPP

APP = TPP / L



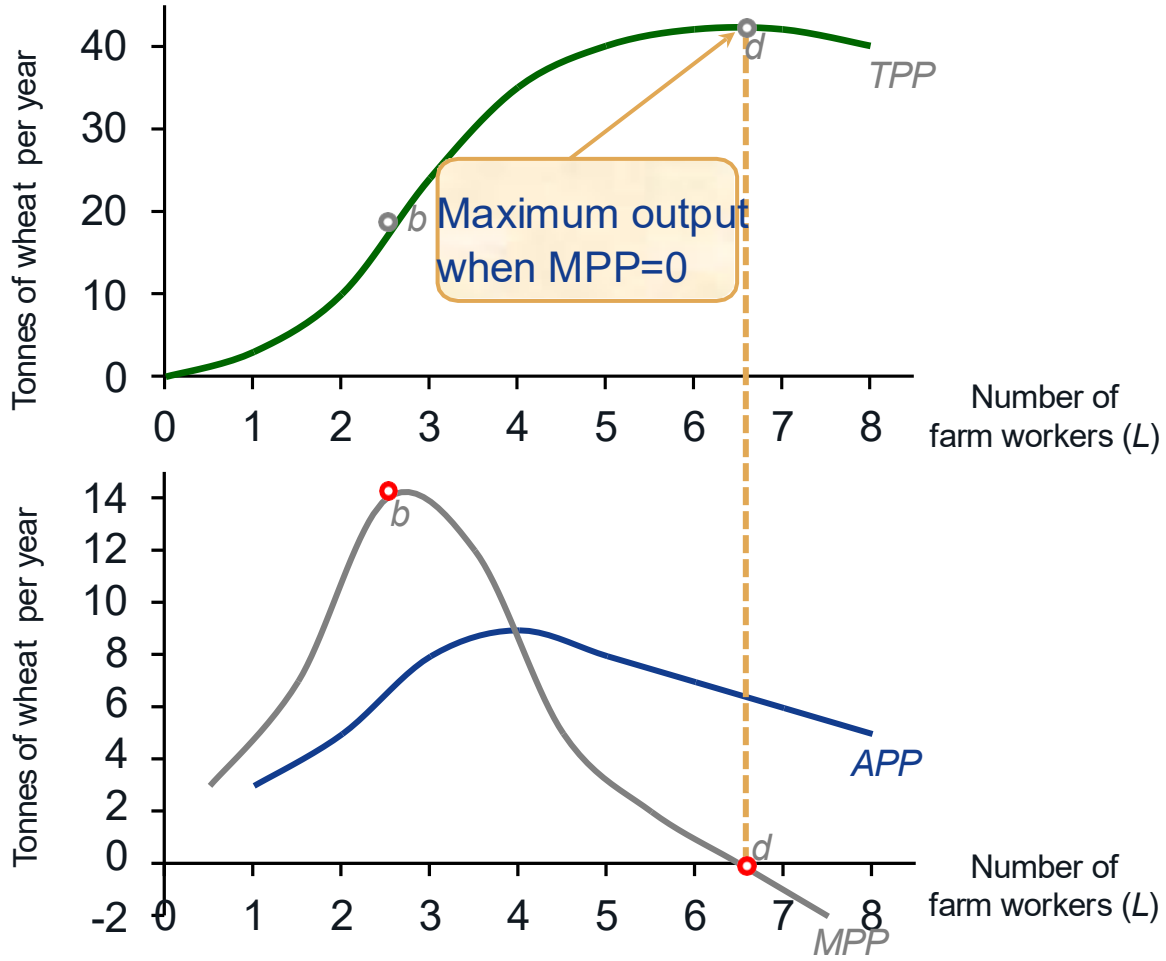
Look at point b.
Up to that point, marginal product is rising.
After point b, marginal product starts to fall.
That is where diminishing marginal returns set in.
Notice again that total product is still increasing after this point.

Farm Production: Maximum TPP and MPP = 0

Total Physical Product (*TPP*)

Marginal Physical Product (*MPP*)

Average Physical Product (*APP*)

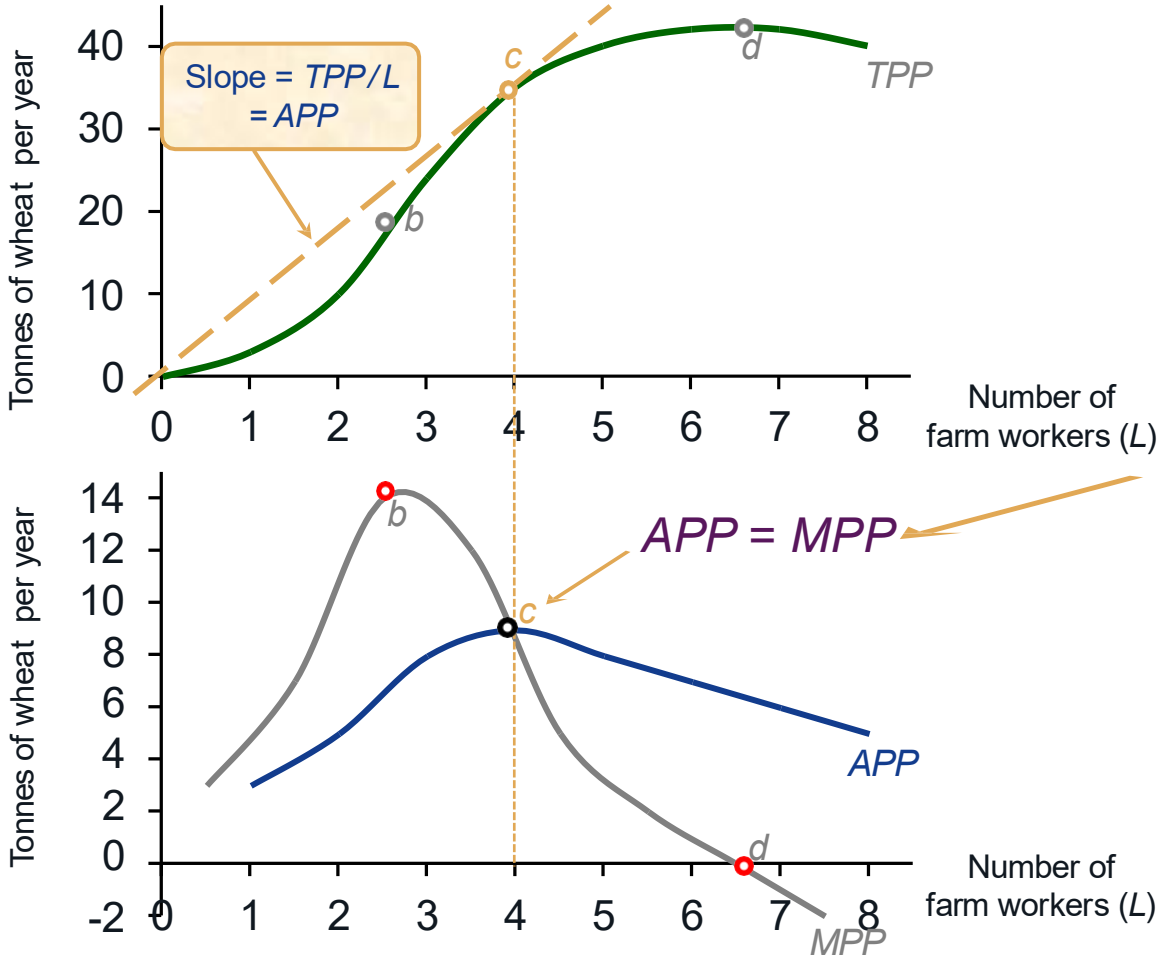


Now look at point d.
At this point, total product reaches its maximum.
Because MPP equals zero. Adding one more worker adds no extra output.
So total output cannot rise any further.
TPP is maximised when MPP equals zero.

Farm Production: Relationship Between APP and MPP

If MPP is above APP, APP rises.
If MPP is below APP, APP falls.
And when MPP equals APP, average product is at its maximum.

Total Physical Product (*TPP*)



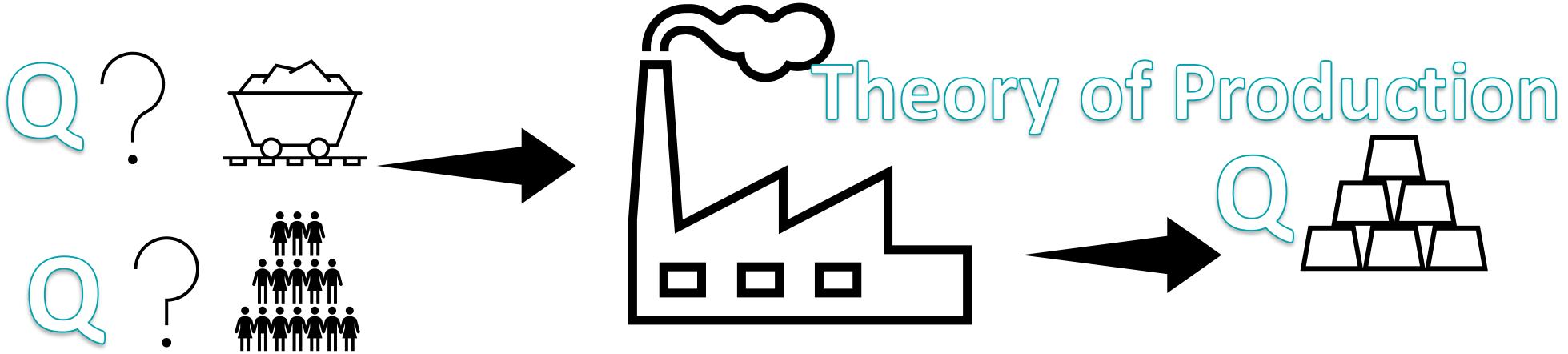
APP is maximised where MPP = APP.

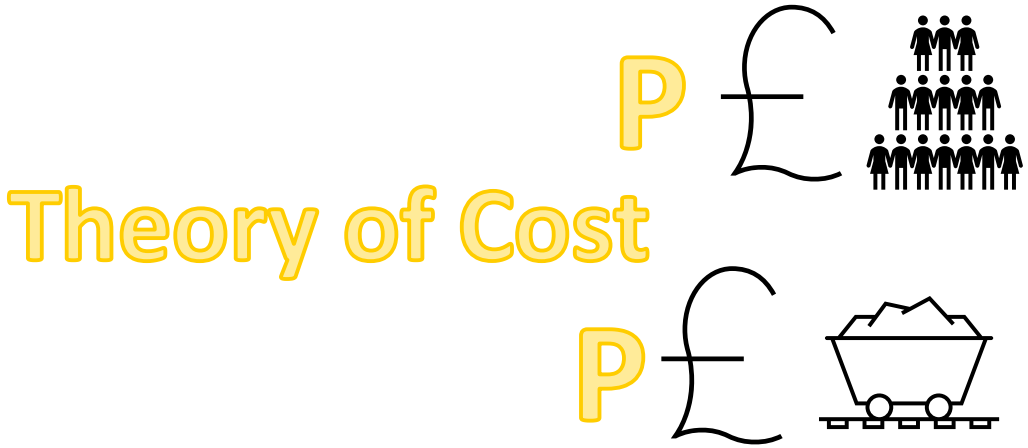
Marginal Physical Product (*MPP*)
the extra output from one additional worker

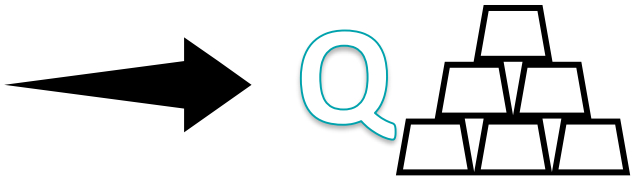
Average Physical Product (*APP*)
average output per worker

Section 2 Background to Supply (Short-run)

2.1 Short-run Costs


Q ?
Q ?
Theory of Production
Q

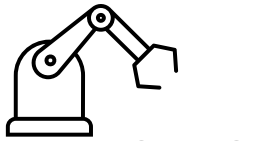

P £
Theory of Cost
P £

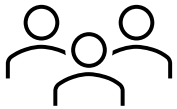

Q


COST

Costs

- Measuring costs of production: opportunity costs
  - **explicit costs**: payments to outside suppliers
  - **implicit costs**: costs that do not involve payment to a third party but involve sacrifice of some alternative(s)
- Fixed costs and variable costs
  - fixed costs and **sunk costs**
- Total costs
  - Total Fixed Cost (TFC)
  - Total Variable Cost (TVC)

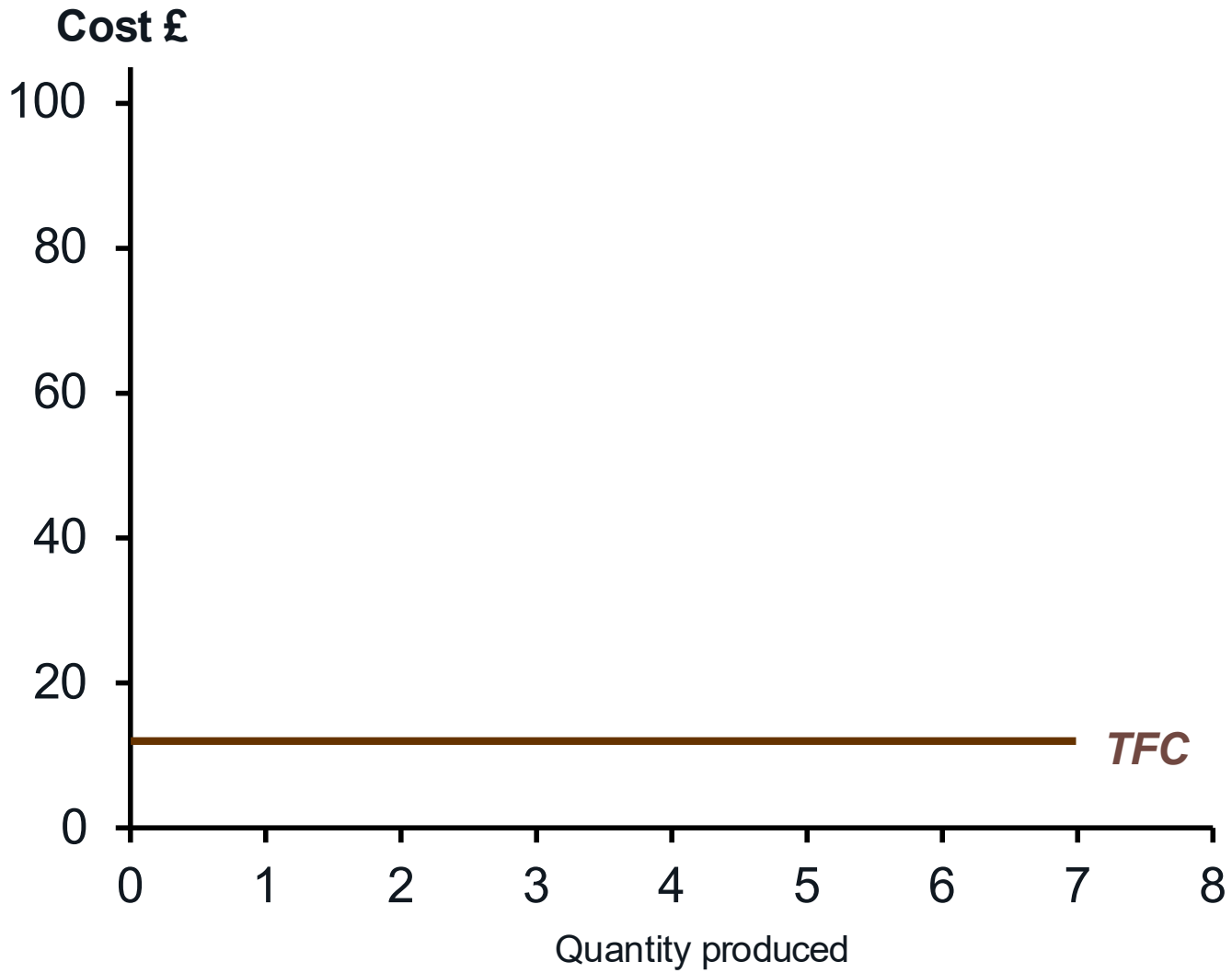




  - Total Cost: $TC = TFC + TVC$ Total Cost =Total Fixed Cost + Total Variable Cost

Total Costs

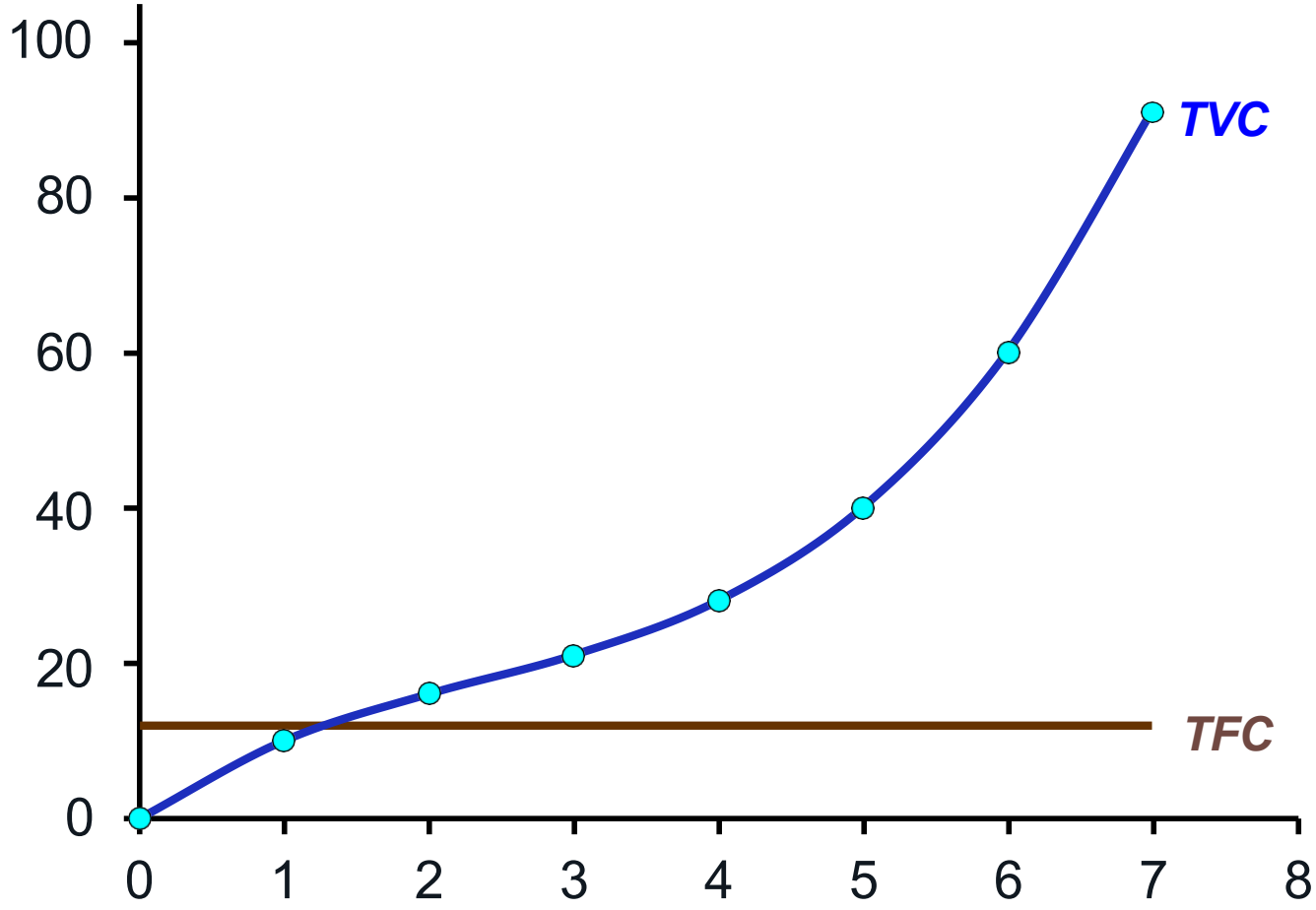
Total Cost (TC) =Total Fixed Cost(TFC) + Total Variable Cost (TVC)



Output (Q)	*TFC* (£)
0	12
1	12
2	12
3	12
4	12
5	12
6	12
7	12

Total Costs

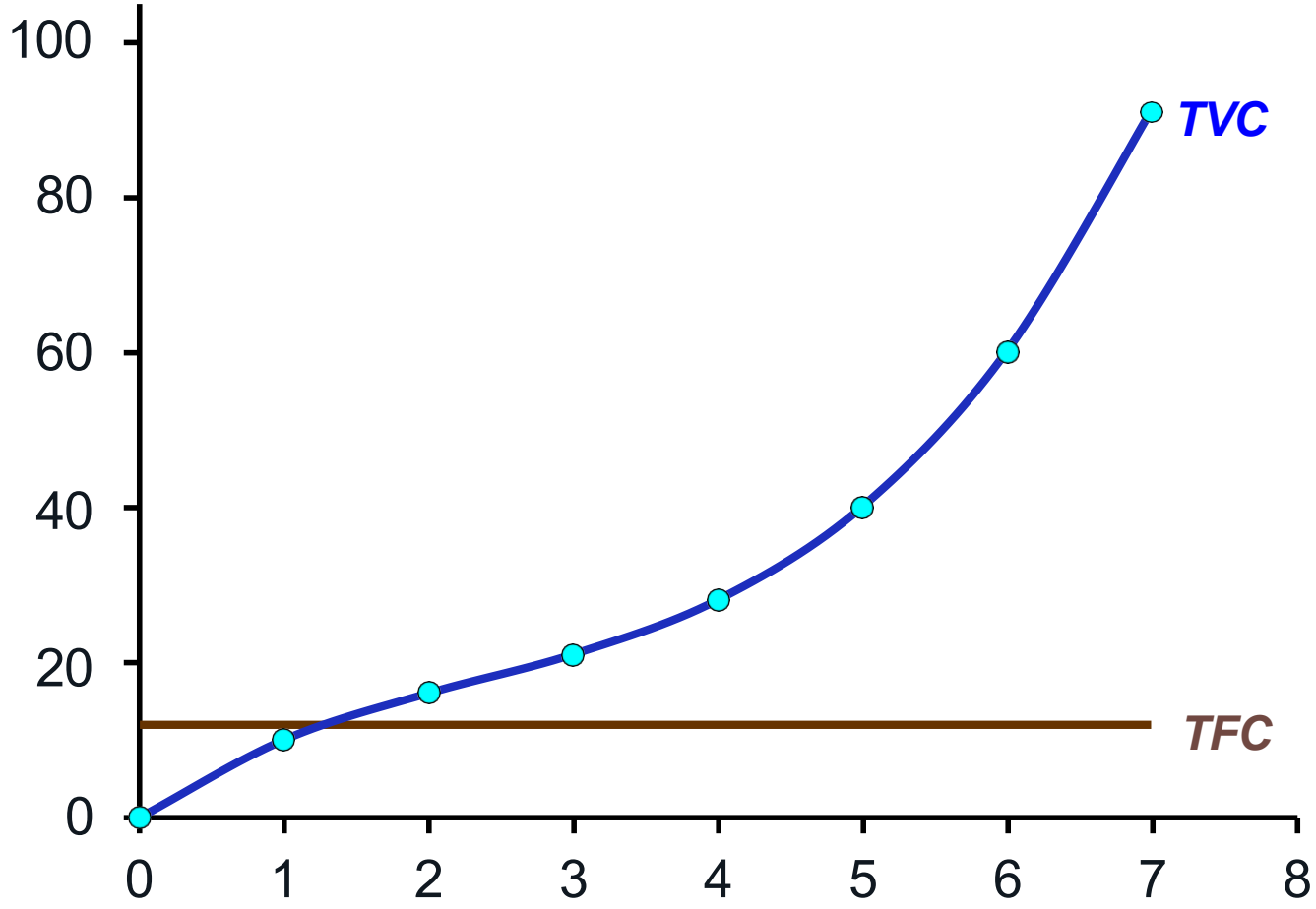
Total Cost (TC) =Total Fixed Cost(TFC) + Total Variable Cost (TVC)



Output (Q)	TFC (£)	TVC (£)
0	12	0
1	12	10
2	12	16
3	12	21
4	12	28
5	12	40
6	12	60
7	12	91

Total Costs

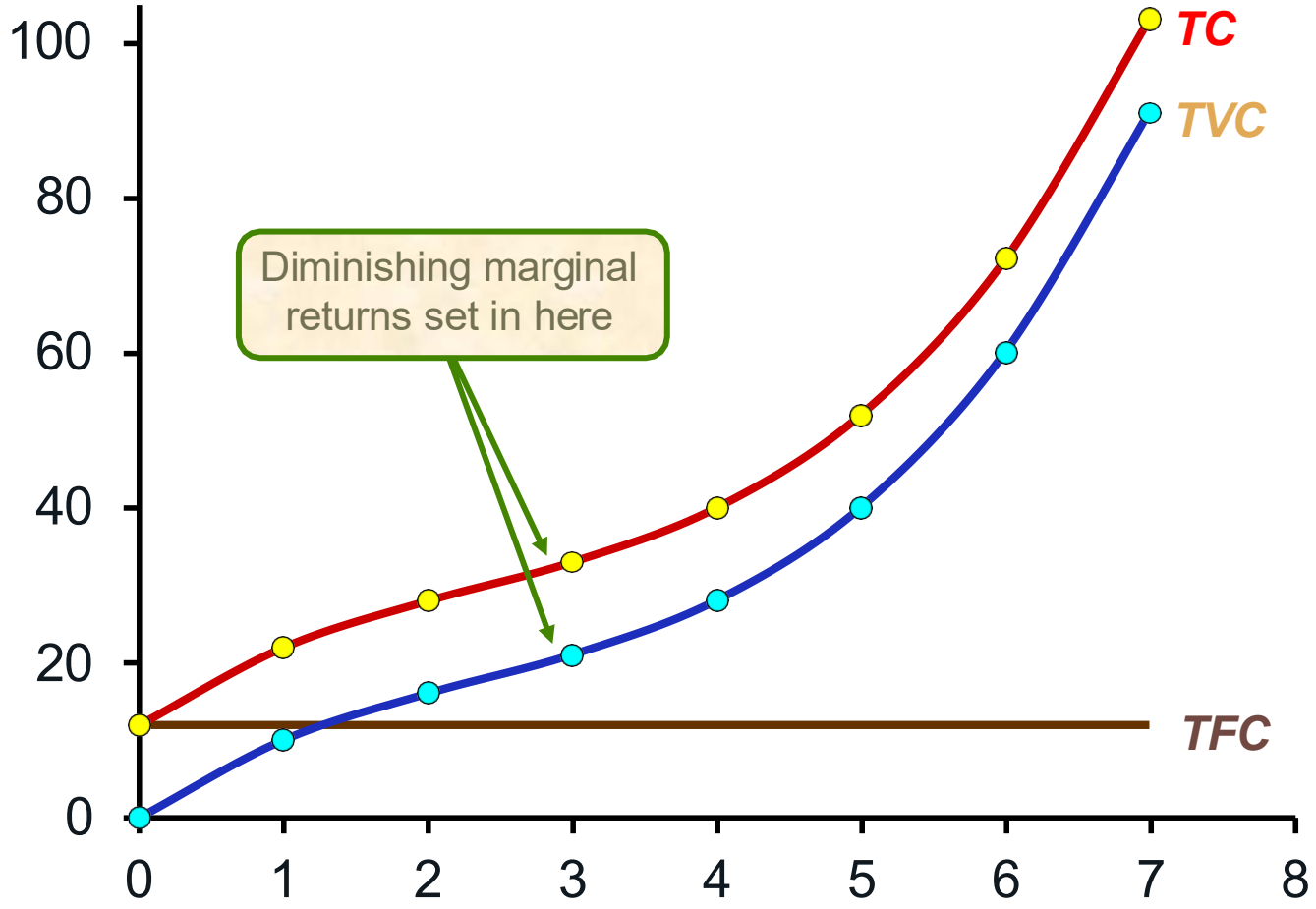
Total Cost (TC) =Total Fixed Cost(TFC) + Total Variable Cost (TVC)



Output (Q)	TFC (£)	TVC (£)	TC (£)
0	12	0	12
1	12	10	22
2	12	16	28
3	12	21	33
4	12	28	40
5	12	40	52
6	12	60	72
7	12	91	103

Total Costs

Total Cost (TC) =Total Fixed Cost(TFC) + Total Variable Cost (TVC)



Output (Q)	TFC (£)	TVC (£)	TC (£)
0	12	0	12
1	12	10	22
2	12	16	28
3	12	21	33
4	12	28	40
5	12	40	52
6	12	60	72
7	12	91	103

Average Cost and Marginal Cost

- **Average Cost: total cost per unit of output.**

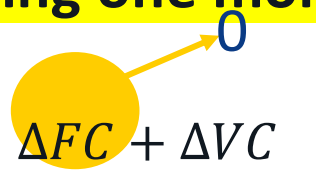
  Average Cost=Average Fixed Cost (AFC) + Average Variable Cost(AVC)

$$AC = \frac{TC}{Q} = \frac{TFC + TVC}{Q} = AFC + AVC$$

  Average Fixed Cost : total fixed cost per unit of output $AFC = \frac{TF}{Q}$

  Average Variable Cost: total variable cost per unit of output $AVC = \frac{\mathrm{TVC}}{\mathrm{Q}}$

- **Marginal cost: the extra cost of producing one more unit of output.**



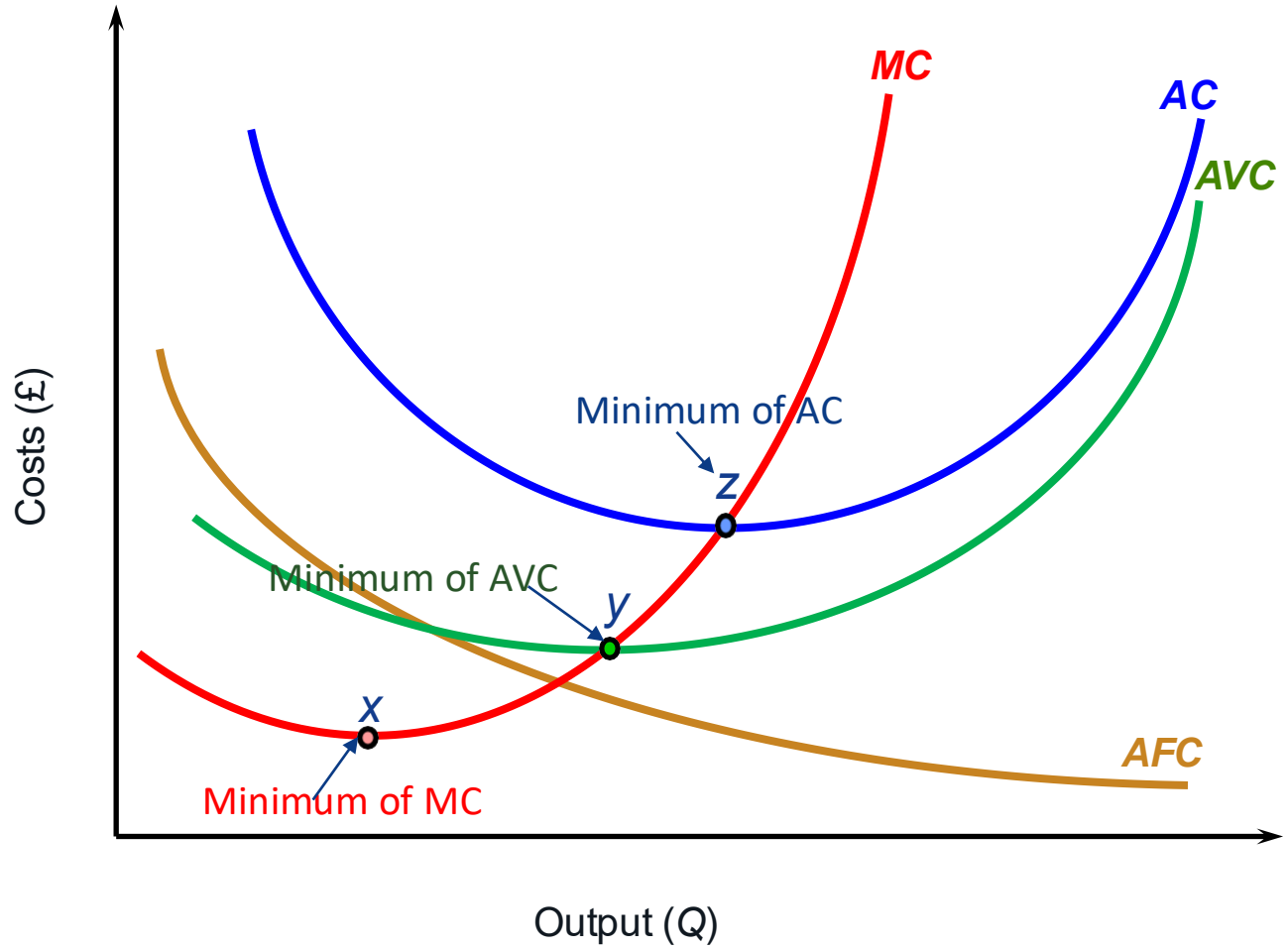
$$MC = \frac{\Delta TC}{\Delta Q} = \frac{\Delta FC + \Delta VC}{\Delta Q} = \frac{\Delta VC}{\Delta Q}$$

  All marginal costs are variable since there are no extra fixed cost as output rises.

Relationship between the marginal and other cost curves

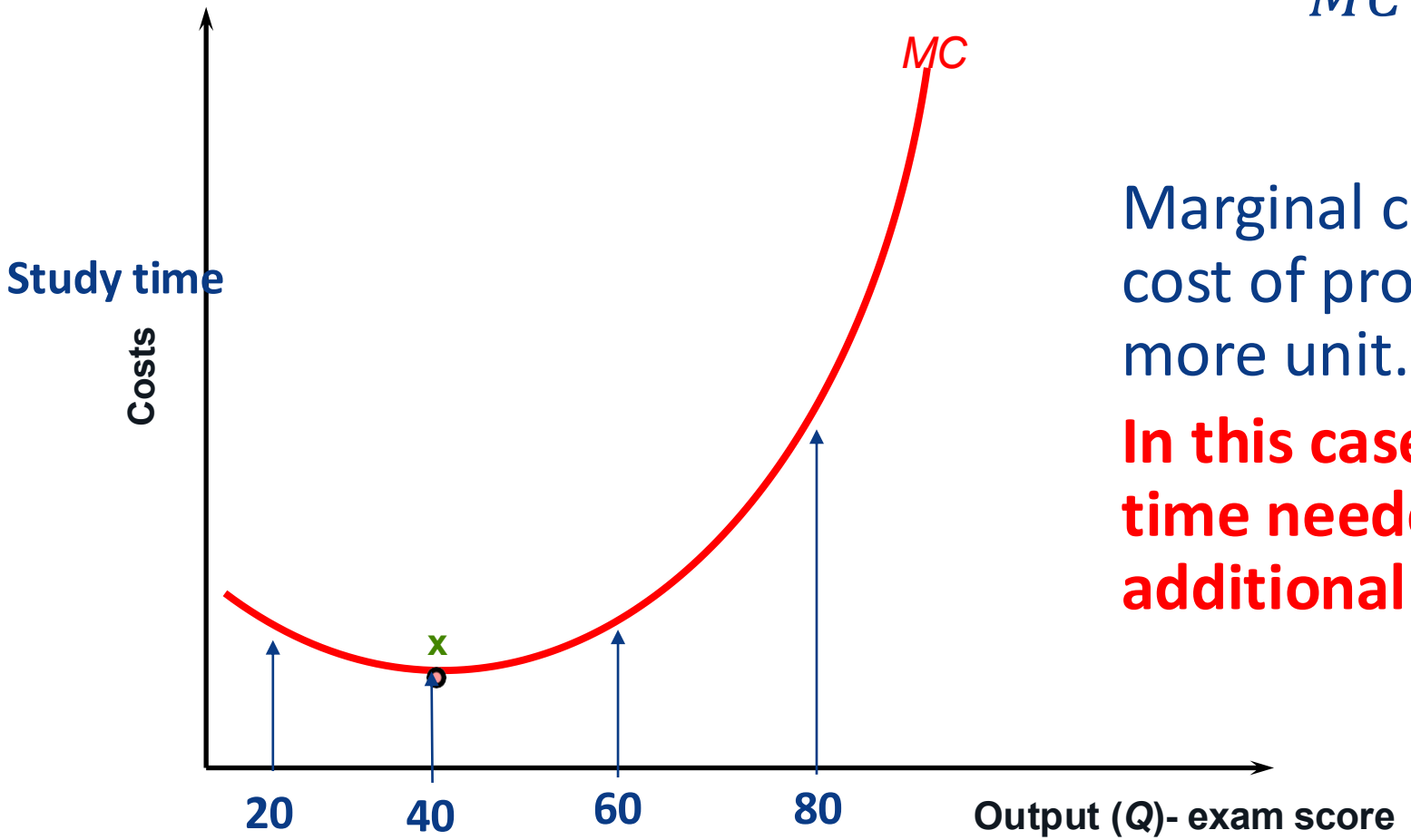
Output (Q) (units)	TFC (£)	AFC (TFC/Q) (£)	TVC (£)	AVC (TVC/Q) (£)	TC (TFC+TVC) (£)	AC (TC/Q) (£)	MC (ΔTC/ΔQ) (£)
0	12	–	0	–	12	–	
							10
1	12	12	10	10	22	22	
							6
2	12	6	16	8	28	14	
							5
3	12	4	21	7	33	11	
							7
4	12	3	28	7	40	10	
							12
5	12	2.4	40	8	52	10.4	
							20
6	12	2	60	10	72	12	
							31
7	12	1.7	91	13	103	14.7	

Relationship between the marginal and other cost curves



Marginal cost- study time and exam mark

$$MC = \frac{\Delta TC}{\Delta Q}$$



Marginal cost: the extra cost of producing one more unit.

In this case: extra study time needed for additional one point

Section 2: Background to Supply (Short-run)

2.2 Short-run Theory of Production and Cost

Relation between marginal cost and marginal physical product

Diminishing marginal returns occur at the point where the marginal physical product (MPP) reaches its maximum and marginal cost (MC) reaches its minimum.

- **Marginal Physical Product (MPP): Change in quantities of output when adding an additional unit of input**

$$MPP = \frac{\Delta TPP}{\Delta Q_L} = \frac{\text{Change In } Quantities\ of\ output\ (TPP)}{\text{Change in Quantities of Input}}$$

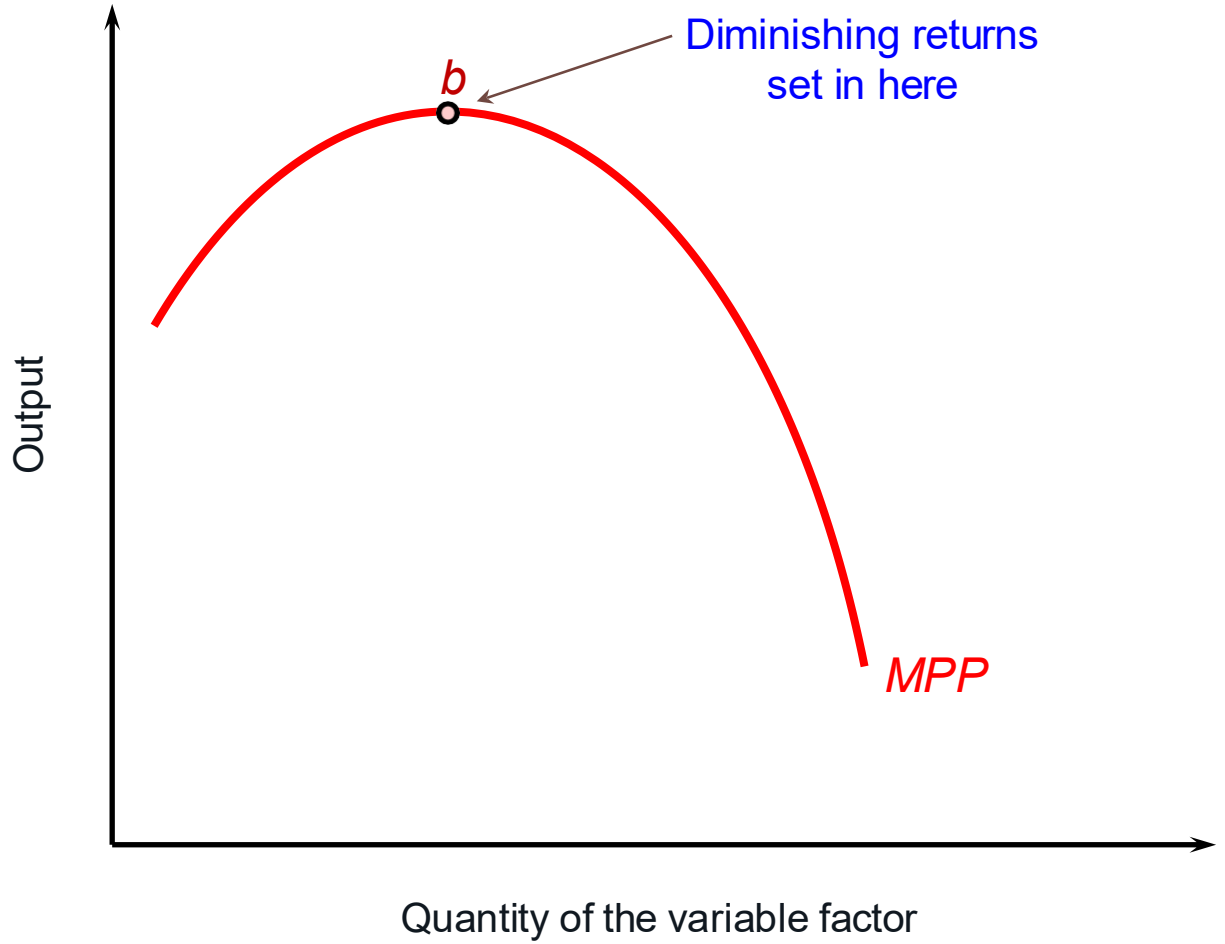
- **Marginal cost: the extra cost of producing one more unit of output**

$$MC = \frac{\Delta TC}{\Delta Q} = \frac{\text{Change In Total Cost}}{Change\ in\ Quantities\ of\ Output(TPP)} = \frac{Change\ in\ VC}{Change\ in\ Quantities\ of\ Output(TPP)}$$

For each additional unit of input (Q input), more quantities of output(TPP) produced, MPP ↑, MC ↓

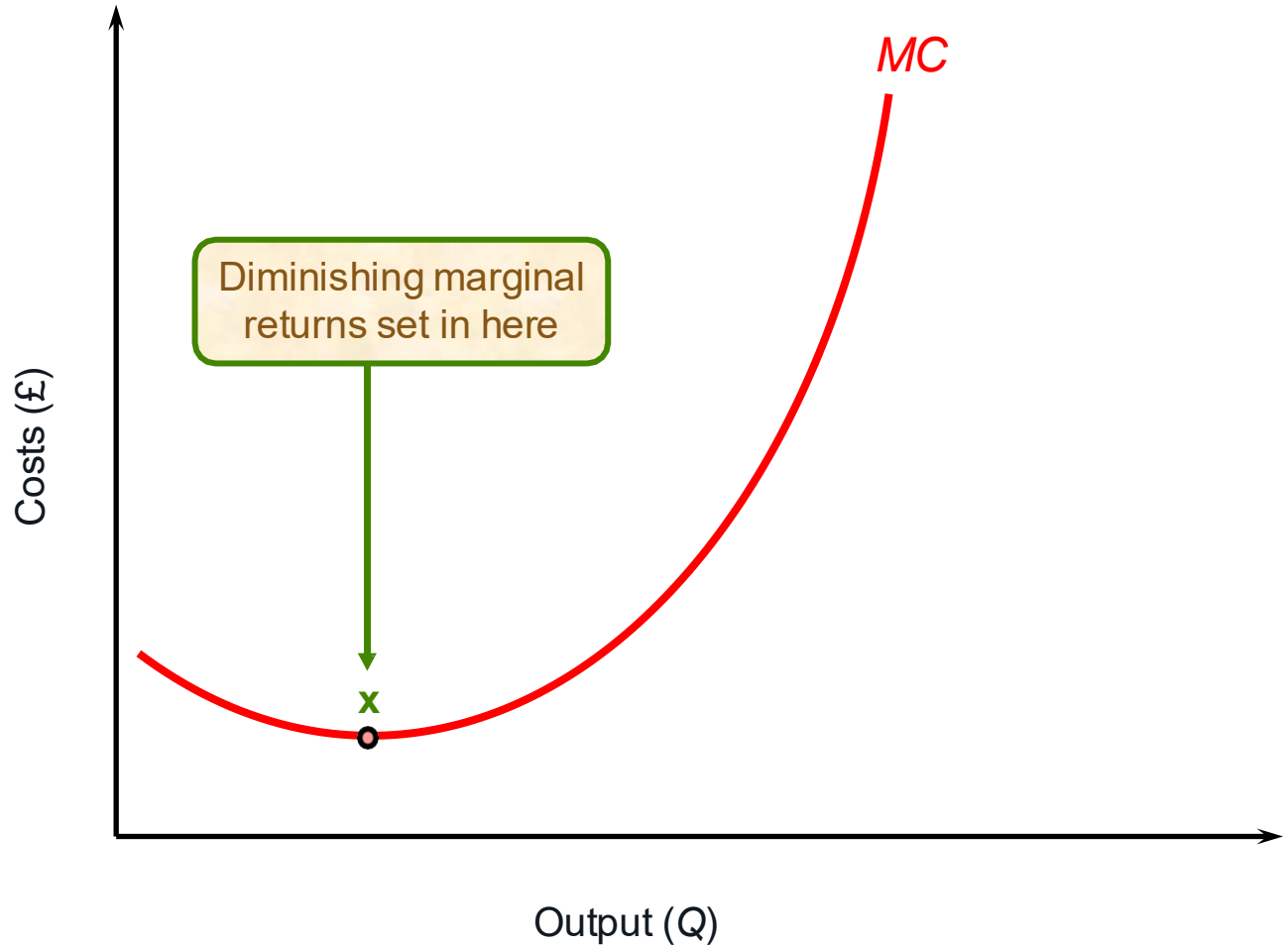
Change in variable cost----- extra cost for an additional unit of input

Marginal Physical Product



Marginal cost

$$MC = \frac{\Delta TC}{\Delta Q}$$



Marginal cost: the extra cost of producing one more unit.

All marginal costs are variable since there are no extra fixed cost as output rises.

Input: Number of Employees	Cost per Person (£)	Total Cost (£)	Output: Total Physical Product (Units)	Marginal Physical Product (MPP)	Marginal Cost (MC) (£)
1	100	100	2	Change in TPP/change in number of employees	Change in total cost/change in Q output
2	100	200	5	3	£100/3 units= 33.33
3	100	300	9	4	£100/4 units =25
4	100	400	14	5	£100/5 units =20
5	100	500	16	2	£100/2 units =50
6	100	600	17	1	£100/1 unit =100
7	100	700	17.5	0.5	£100/0.5 units =200

Marginal Physical Product (MPP): Change in quantities of output when adding an additional unit of input $MPP = \frac{\Delta TPP}{\Delta Q_L} = \frac{\text{Change In } Quantities\ of\ output\ (TPP)}{\text{Change in Quantities of Input (\# of employees)}}$

Marginal cost: the extra cost of producing one more unit of output

$$MC = \frac{\Delta TC}{\Delta Q} = \frac{\text{Change In Total Cost}}{Change\ in\ Quantities\ of\ Output(TPP)} = \frac{Change\ in\ VC}{Change\ in\ Quantities\ of\ Output(TPP)}$$

Section 3: Background to Supply (Long run)

3.1 Long-run Theory of Production

Recall:

Production function $Q = f(L, K)$

Where Q represents the total quantity of output, L is labor, and K is capital.

- **Short-run:**

  K (Fixed input) remains constant e.g. machinery

  L increases, Q increases first then decrease (**law of diminishing marginal returns**)

- **Long run:**

  L, K are both variable

  Economies of scale/ of scope



Image source: Getty Images/iStockphoto

Long-run Theory of Production and Cost

In the long run, all factors of production are variable, the costs are variable as well.

In the long run, firms aim to **optimize their production processes** by adjusting input levels to achieve the most efficient and cost-effective production.

- Scale of production
- Production technology



Long-run Theory of Production

The scale of production: how much to produce (quantities of output)

The relationship between inputs (Quantities) and outputs (Quantities)

- **Increasing returns**

  x% increase in inputs leads to an (x+y)% increase in output

- **Constant returns**

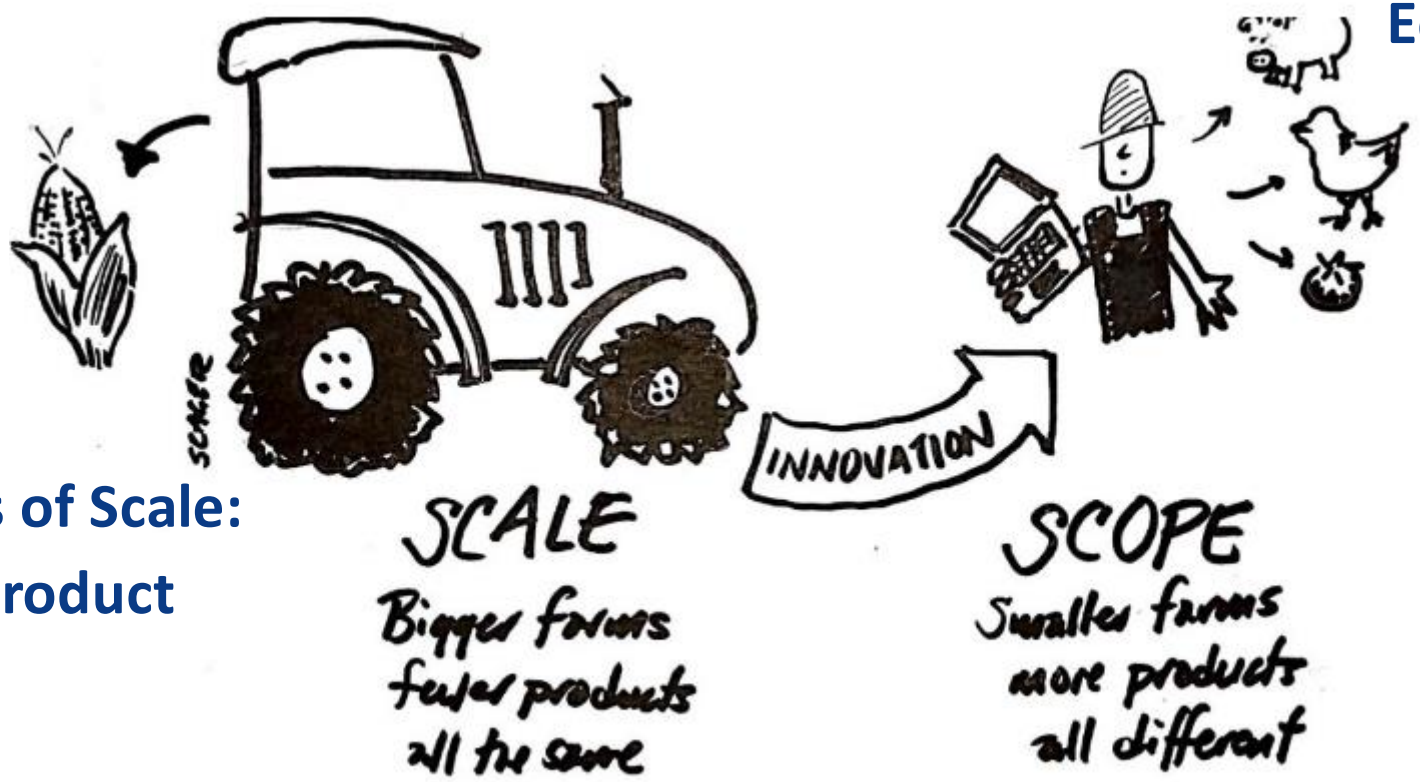
  x% increase in inputs leads to an x% increase in output

- **Decreasing returns**

  x% increase in inputs leads to an (x-y)% increase in output

Long-run Theory of Production

Economies of scale VS. Economics of scope



Economics of Scope: Multiple Products

Economics of Scale: Single Product

Image source from: https://www.linkedin.com/pulse/economies-scale-vs-scope-apexhatchers

Long-run Theory of Production

Economies of scale

When increasing the scale of production leads to a lower cost per unit of output

- Specialisation of labour
- Indivisibilities
- Efficiency of large plant
- The “container principle”
- Multi-stage production
- Finance
- Organisation



Long-run Theory of Production

Economies of scope

When increasing the range of products produced by a firm reduces the cost of producing each one

- Sharing of overheads between product lines

Cost Advantages for a range of products!



Optimal combination of inputs (factors)-Productive efficiency

- **two-factor case**
  - If $\frac{MPP_a}{P_a} > \frac{MPP_b}{P_b}$, costs can be reduced by using more factor a and less factor b
  - costs minimised where:

$$\frac{MPP_a}{P_a} = \frac{MPP_b}{P_b}$$

- **multi-factor case**
  - costs minimised where:

$$\frac{MPP_a}{P_a} = \frac{MPP_b}{P_b} = \cdots = \frac{MPP_n}{P_n}$$

(*equi-marginal principle*)

Marginal Physical Product (MPP): Incremental impact of adding an additional unit of **input** on the quantity of outputs

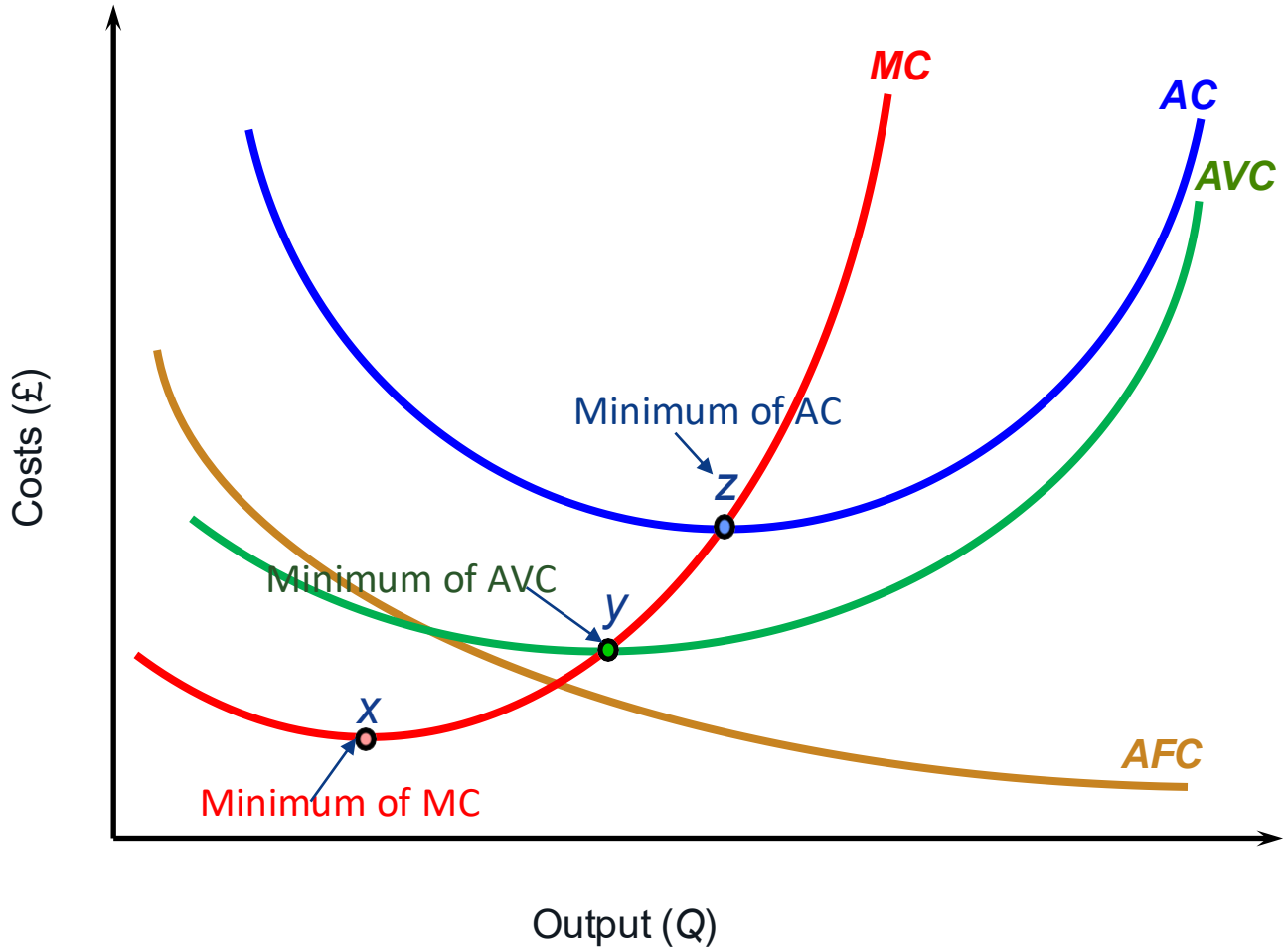
P is the price of factors (cost per unit of input)

MPP/P ---- productivity per $(£)

Section 3 Background to Supply (Long run)

3.2 Long-run Theory of Cost

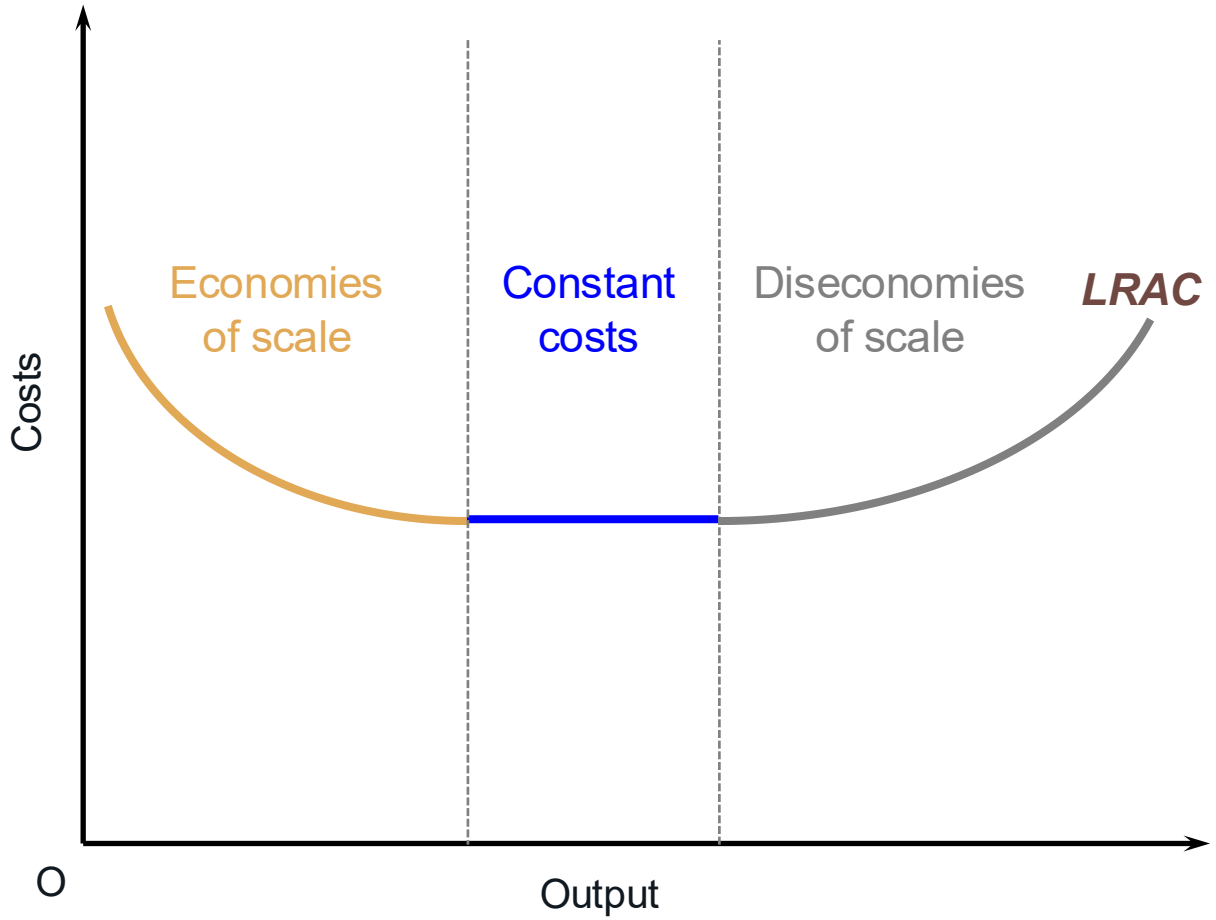
Recall: Costs in the short run



Typical long-run average cost curve

Long-run average cost (LRAC):

total cost per unit of output in the long run



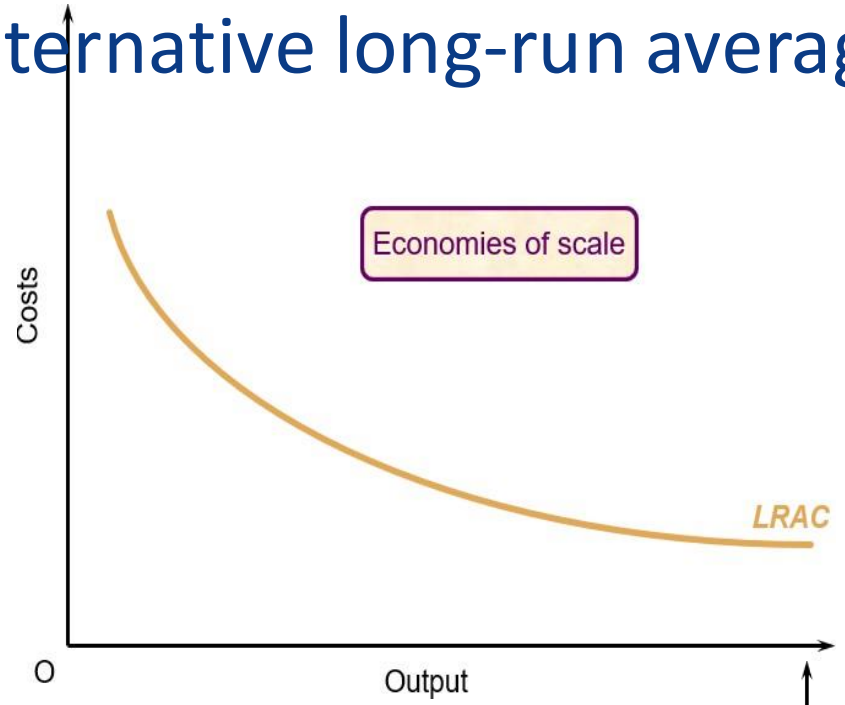
Economies of scale:

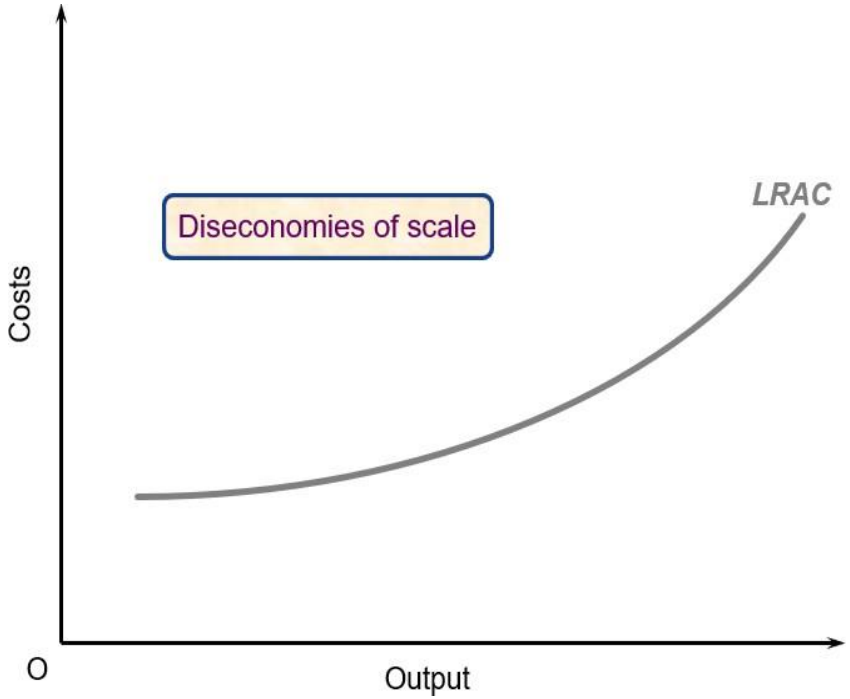
When increasing the scale of production leads to **a lower** cost per unit of output

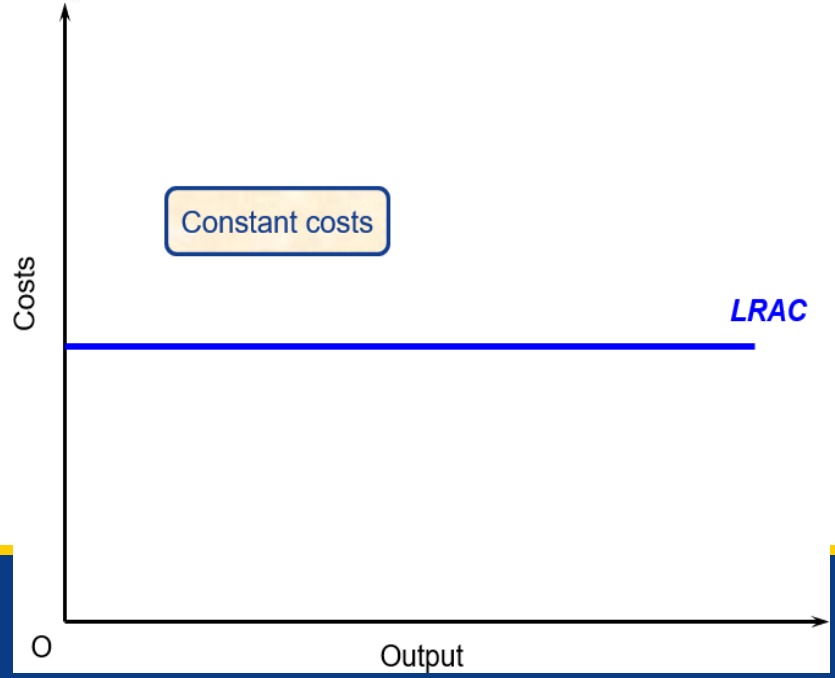
Diseconomies of scale:

When increasing the scale of production, the cost per unit of output **increases**.

Alternative long-run average cost curves







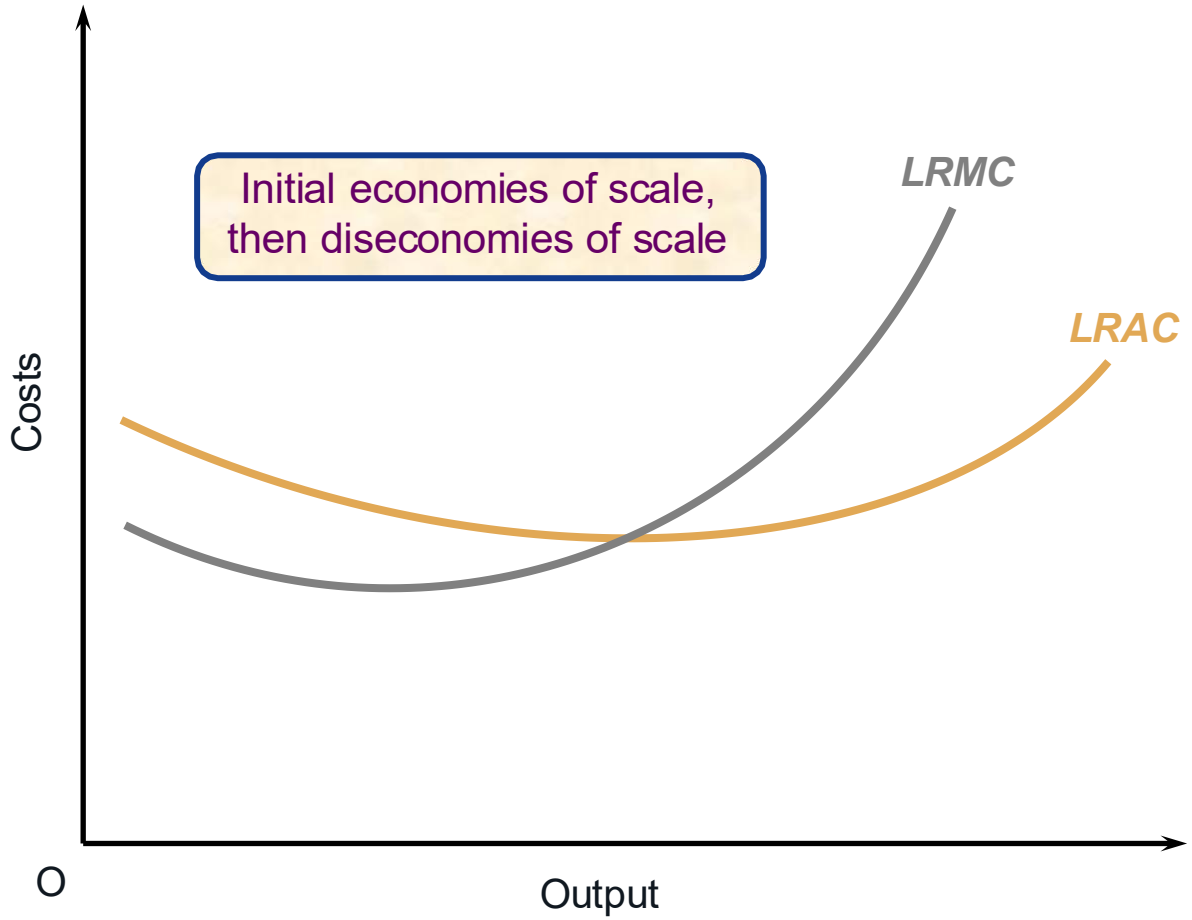
Long-run average and marginal cost curves

Long-run average cost (LRAC):

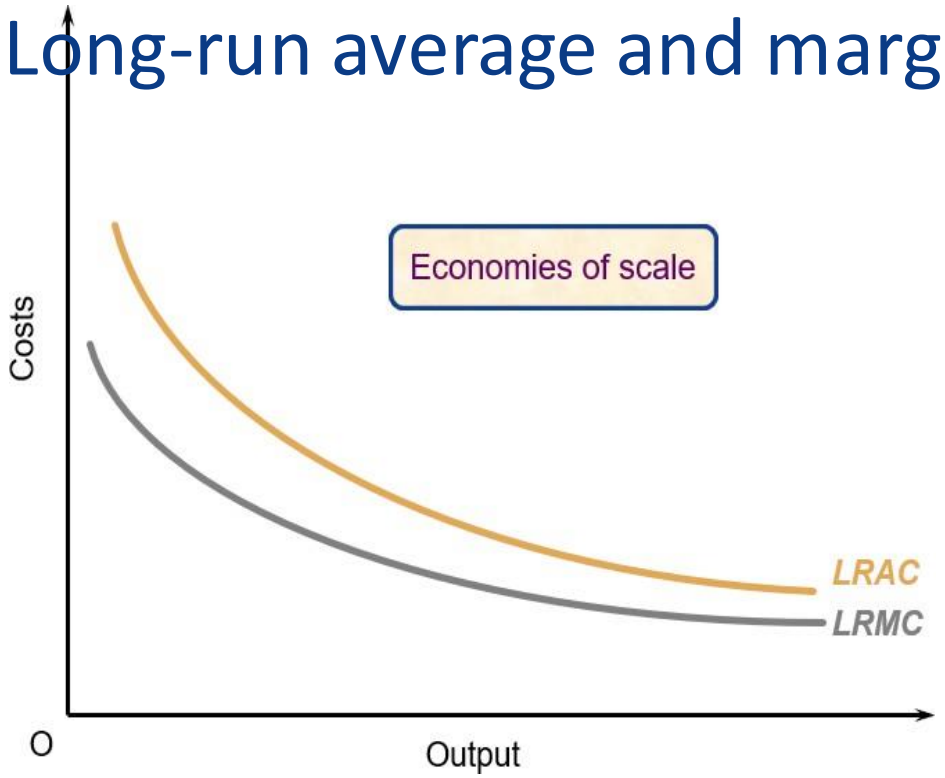
total cost per unit of output in the long run

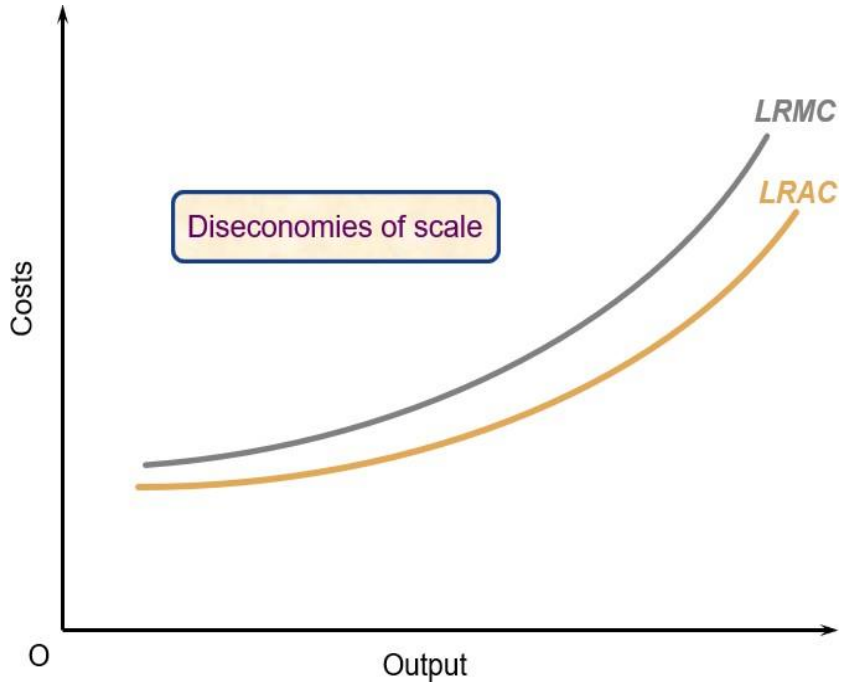
Long-run marginal cost (LRMC):

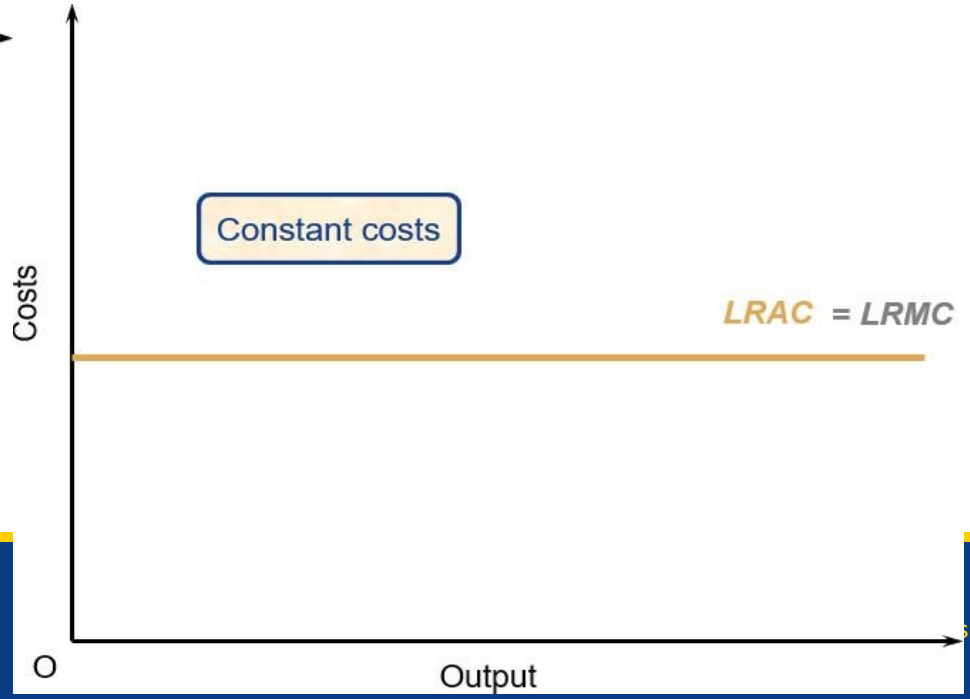
extra cost of producing one additional unit of output in the long run



Long-run average and marginal cost curves







Section 4 Background to Supply

4.1 Revenue: Intro

Revenue



Total Revenue (TR)

- The total amount of money a firm earns from selling its products or services over a specific period.
- TR = Price (P) x Quantity Sold (Q)

Average Revenue (AR)

- As known as Price per unit, represents the revenue generated by each unit of the product sold.
- AR = TR / Q

Marginal Revenue (MR)

- the additional revenue a firm earns by selling one more unit of its product or service.
- MR = ΔTR / ΔQ

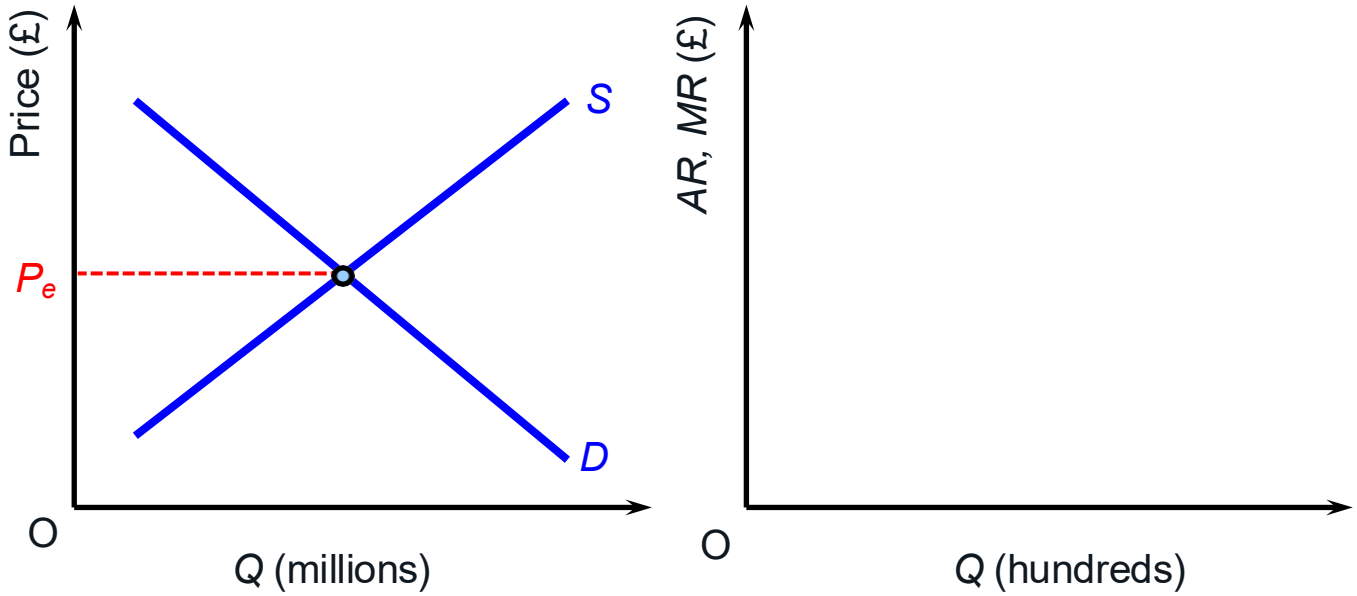
Section 4 Background to Supply

4.2 Revenue when firms are price-takers (Perfectly competitive market)

Deriving a firm's AR and MR: Price-taker

Demand and Supply

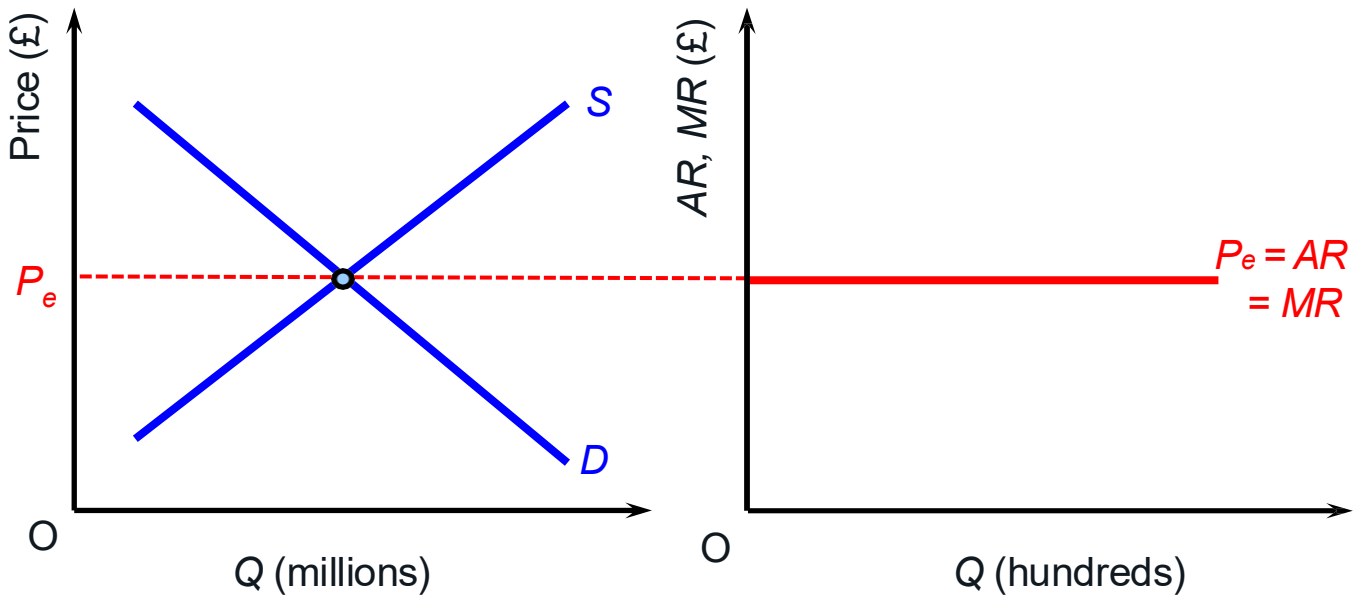
Assumption : Perfectly competitive market (the same products; many sellers)



(a) The market **(b)** The firm

Deriving a firm's AR and MR: Price-taker

Market equilibrium determine the selling price due to price taker



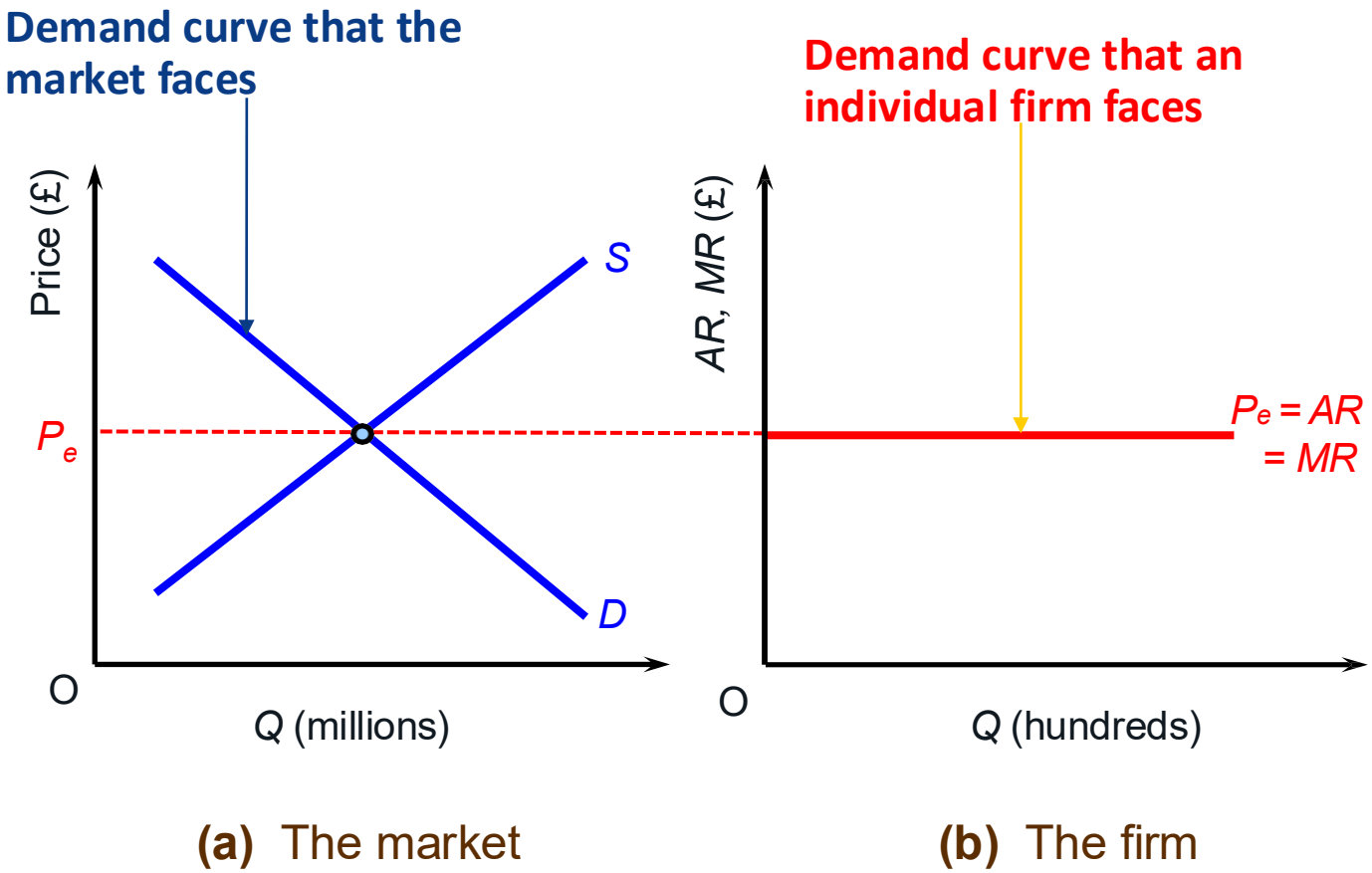
(a) The market **(b)** The firm

Marginal Revenue (MR): the additional revenue a firm earns by selling one more unit of its product

Additional revenue for one more unit of product (MR)= Price=AR

Average Revenue (AR): as known as price per unit

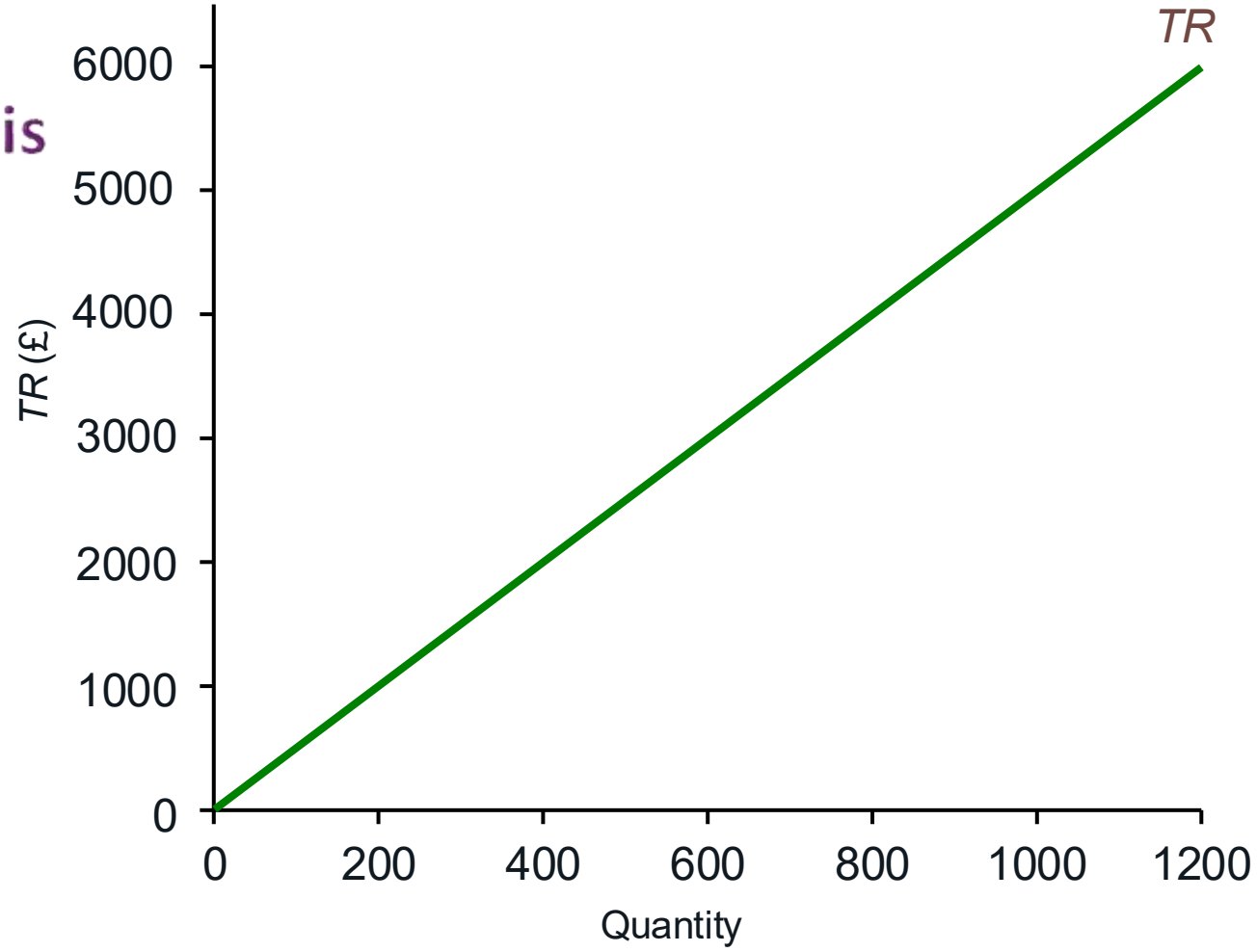
Deriving a firm's AR and MR: Price-taker



Total revenue: Price-taker

Total Revenue(TR) = Price (P) x Quantity Sold (Q)

Price taker: price is the same regardless of quantities sold



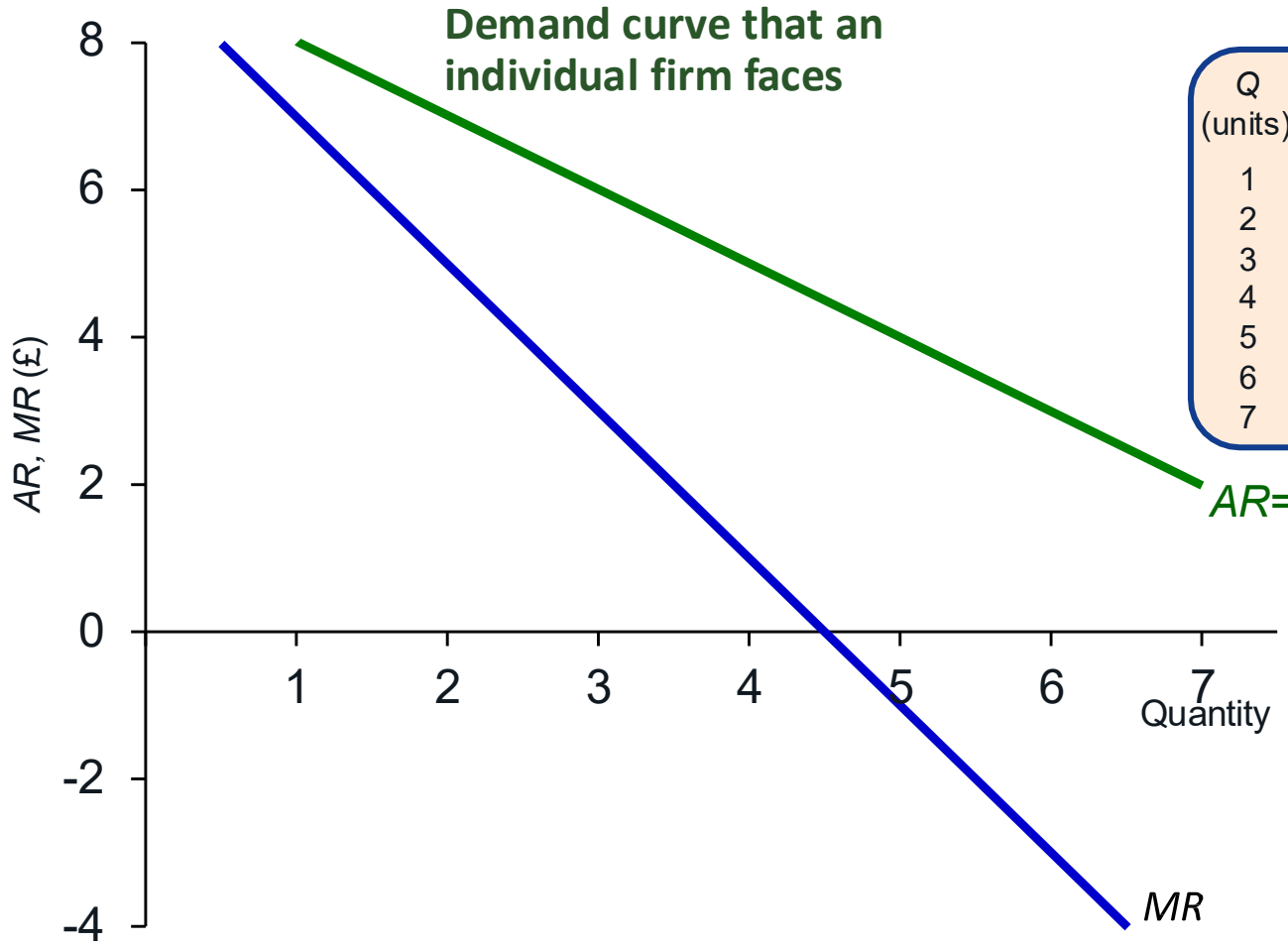
Quantity (units)	Price = *AR* = *MR* (£)	*TR* (£)
0	5	0
200	5	1000
400	5	2000
600	5	3000
800	5	4000
1000	5	5000
1200	5	6000

Section 4 Background to Supply

4.3 Revenue in alternative markets

AR and MR curves for a firm facing downward- sloping demand curve

Price reduces when quantities sold increases



Q (units)	P =AR (£)	TR (£)	MR (£)
1	8	8	
2	7	14	6
3	6	18	4
4	5	20	2
5	4	20	0
6	3	18	–2
7	2	14	–4

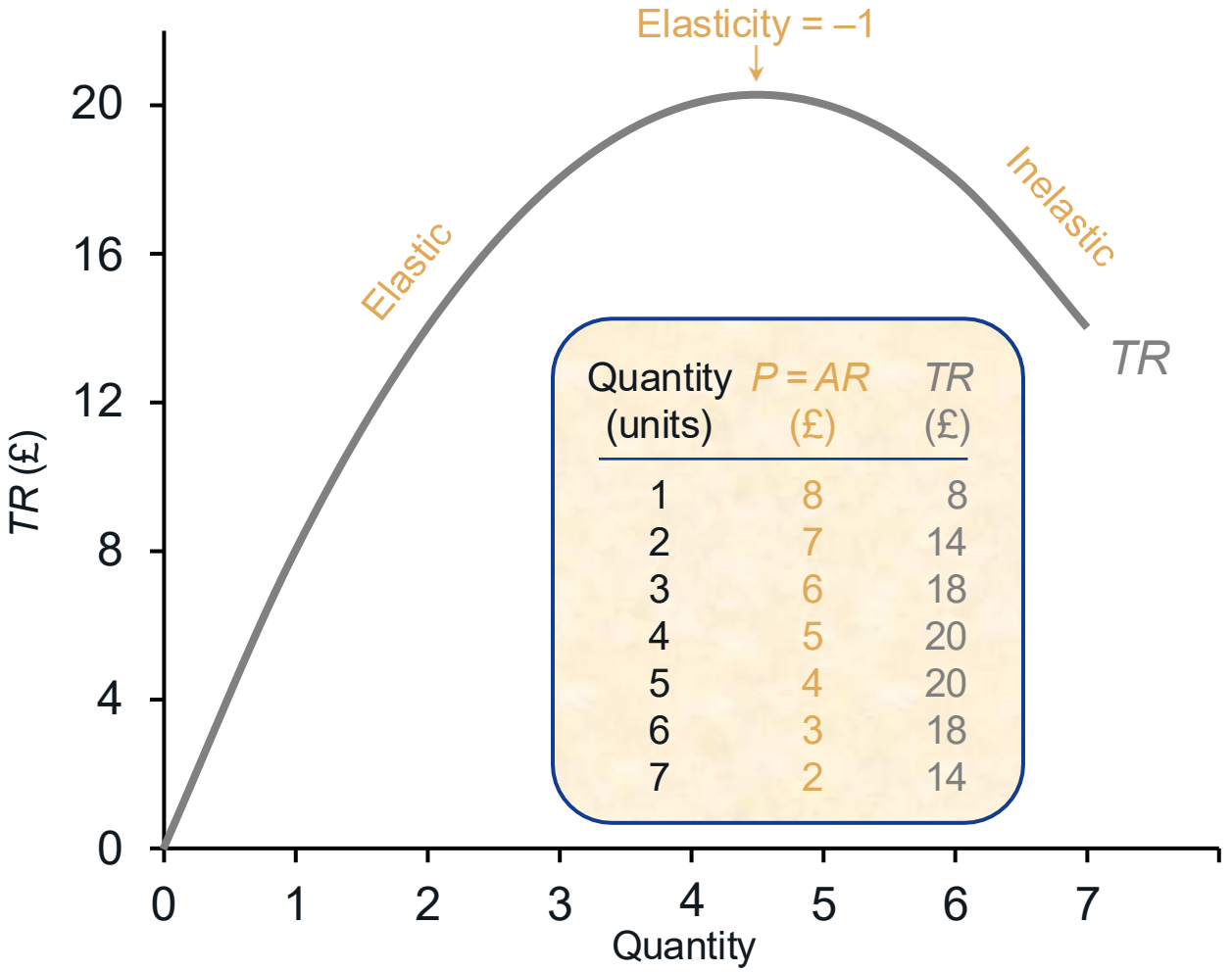
Average Revenue (AR): as known as price per unit

Marginal Revenue (MR): the additional revenue a firm earns by selling one more unit of its product

$MR = \Delta TR / \Delta Q$

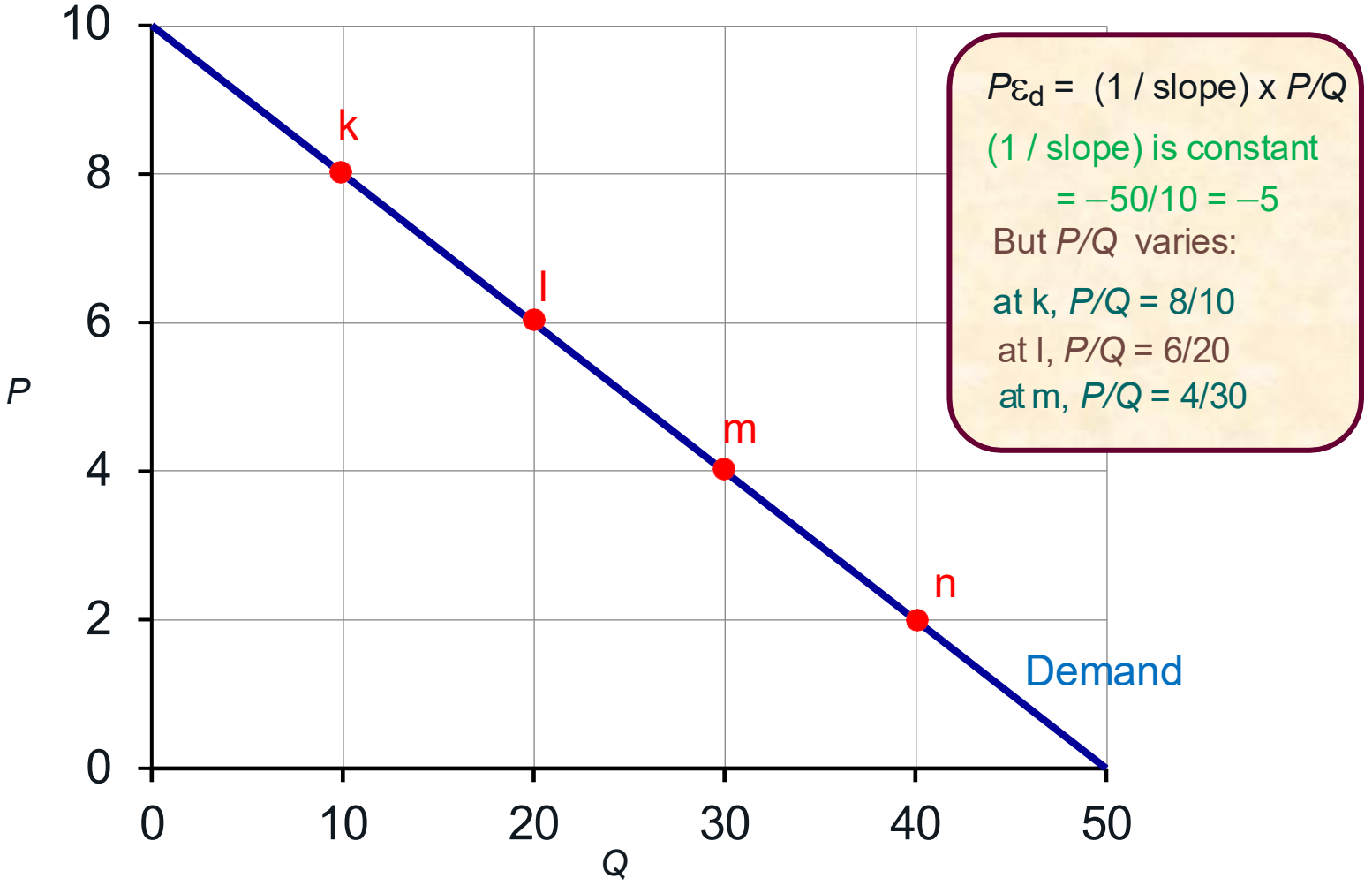
TR curve for a firm facing downward-sloping demand curve

Price reduces when quantities sold increases



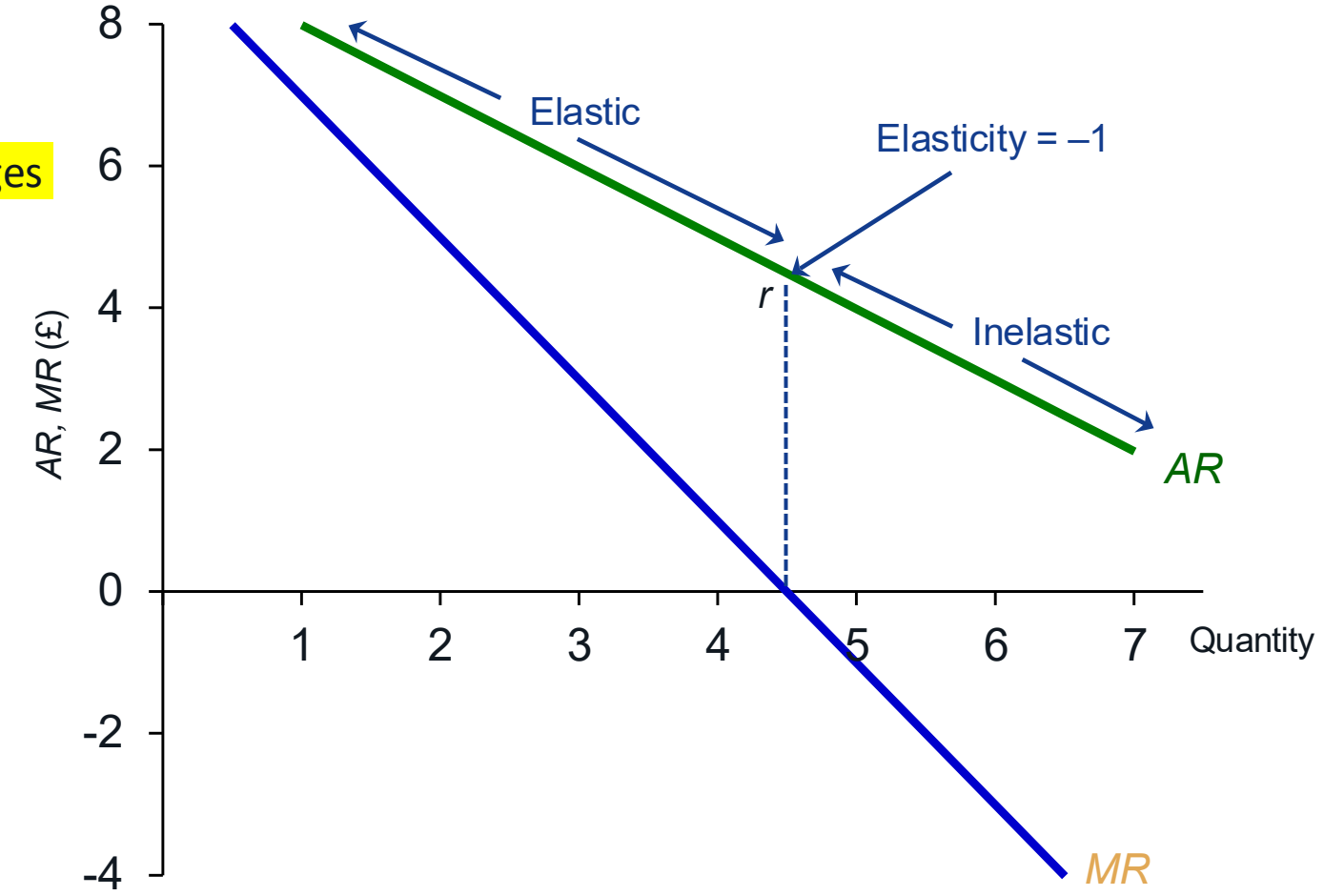
Quantity (units)	P = AR (£)	TR (£)
1	8	8
2	7	14
3	6	18
4	5	20
5	4	20
6	3	18
7	2	14

Recall: Lecture 2
Different elasticities along straight-line demand curve



AR and MR curves for a firm facing downward- sloping demand curve

- **Elasticity of demand** measures of ***how much*** buyers respond to changes in price.



Section 5: Background to Supply

Profit Maximisation

Deriving total profit

Profit:

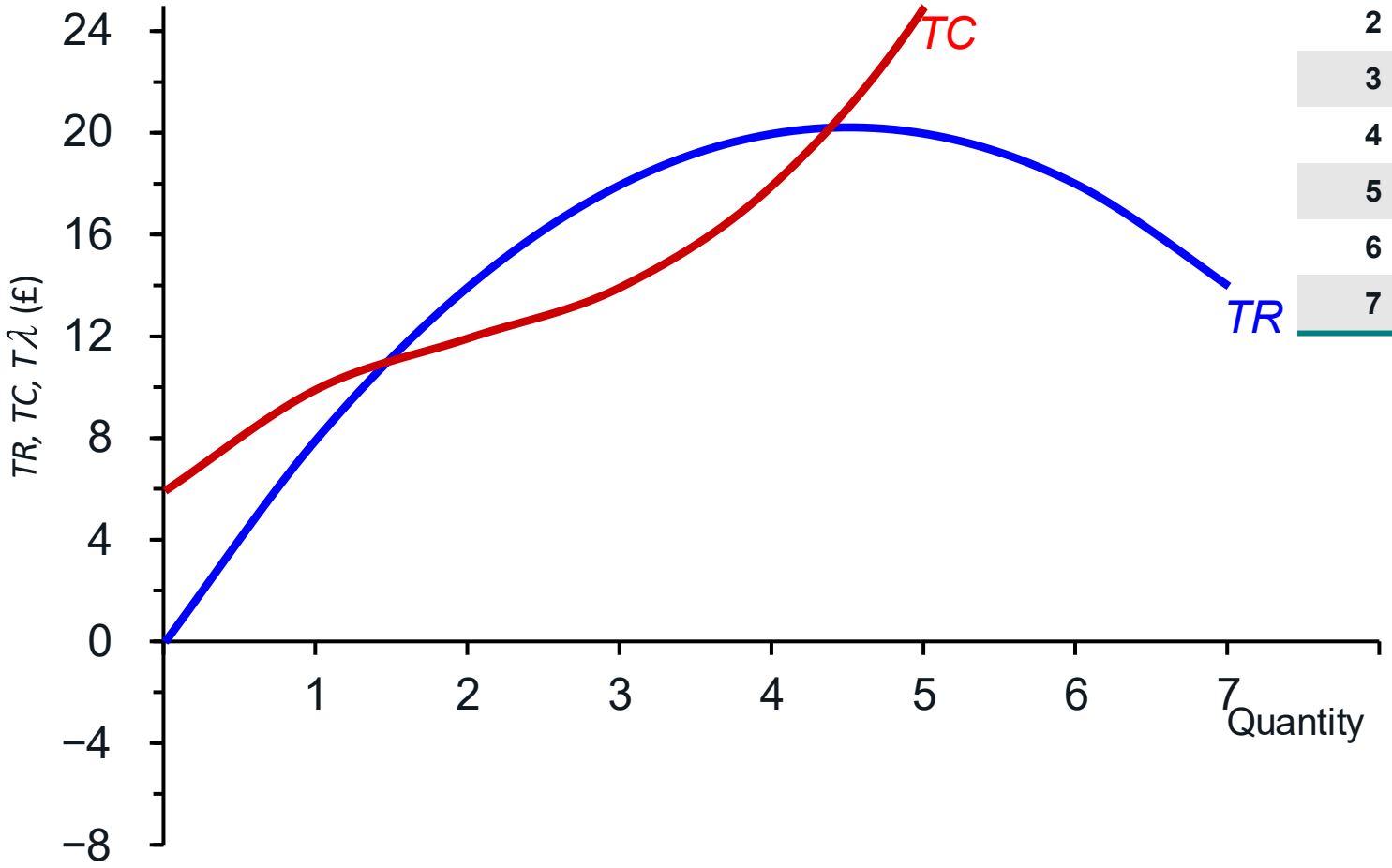
$$Profit(T\Pi) = Total\ Revenue(TR) - Total\ Cost\ (TC)$$



Q (units)	TR (£)	TC (£)	TΠ (£)
0	0	6	–6
1	8	10	–2
2	14	12	2
3	18	14	4
4	20	18	2
5	20	25	–5
6	18	36	–18
7	14	56	–42

Maximum profit using total curves

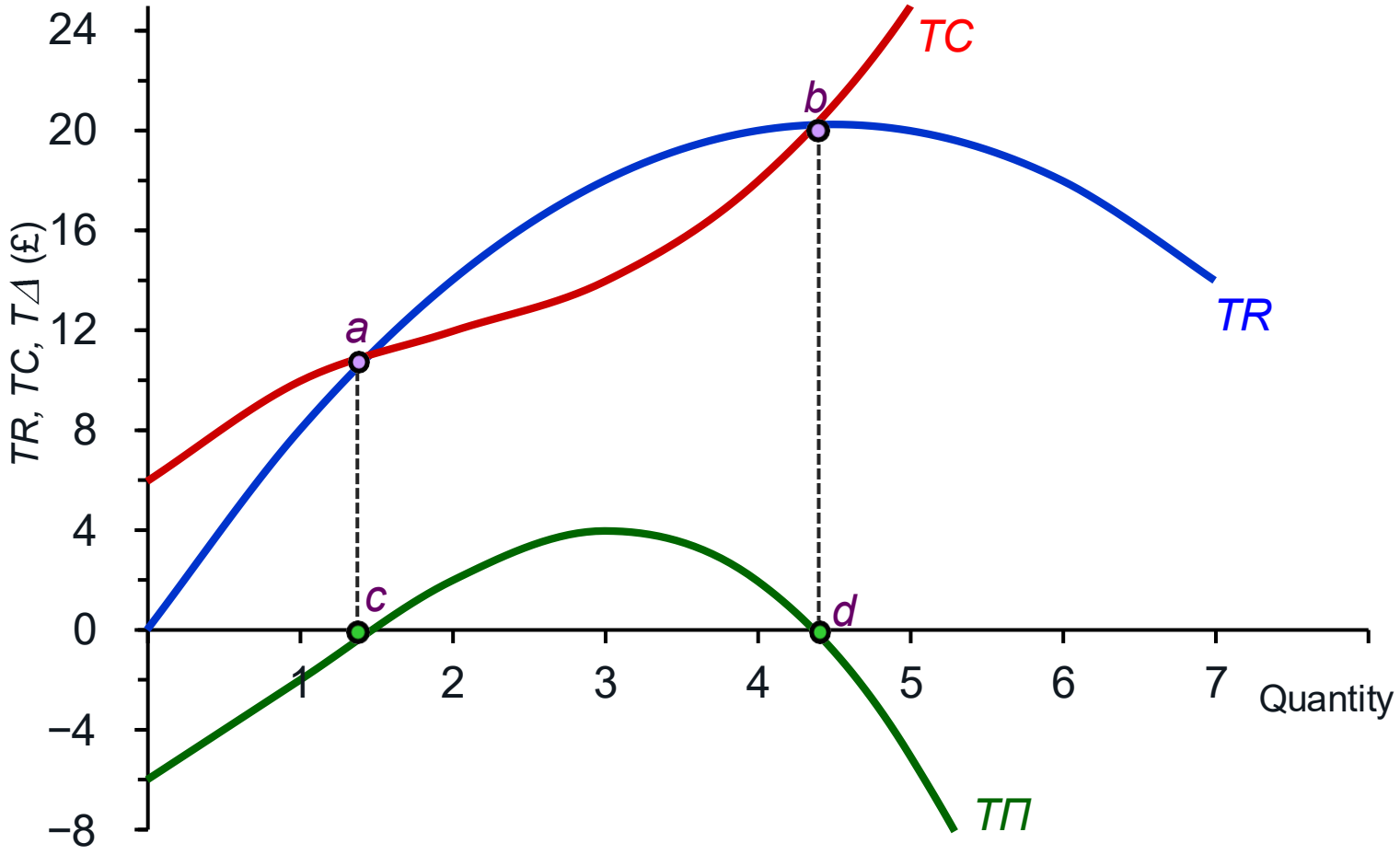
Q (units)	TR (£)	TC (£)	TΠ (£)
0	0	6	–6
1	8	10	–2
2	14	12	2
3	18	14	4
4	20	18	2
5	20	25	–5
6	18	36	–18
7	14	56	–42



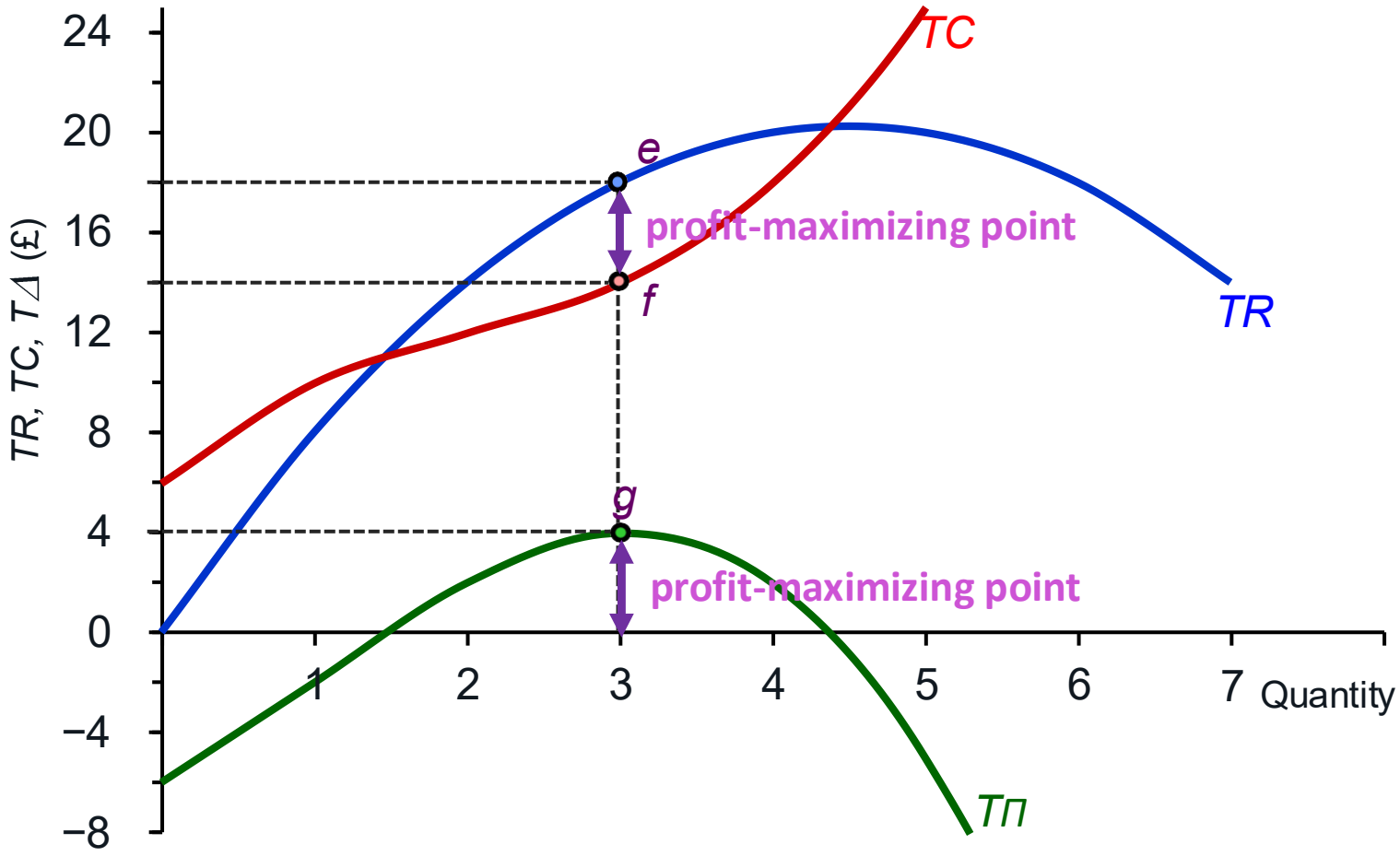
Maximum profit using total curves

Profit:

$$Profit(T\Pi) = Total\ Revenue(TR) - Total\ Cost\ (TC)$$



Maximum profit using total curves



Profit maximisation using marginal curves

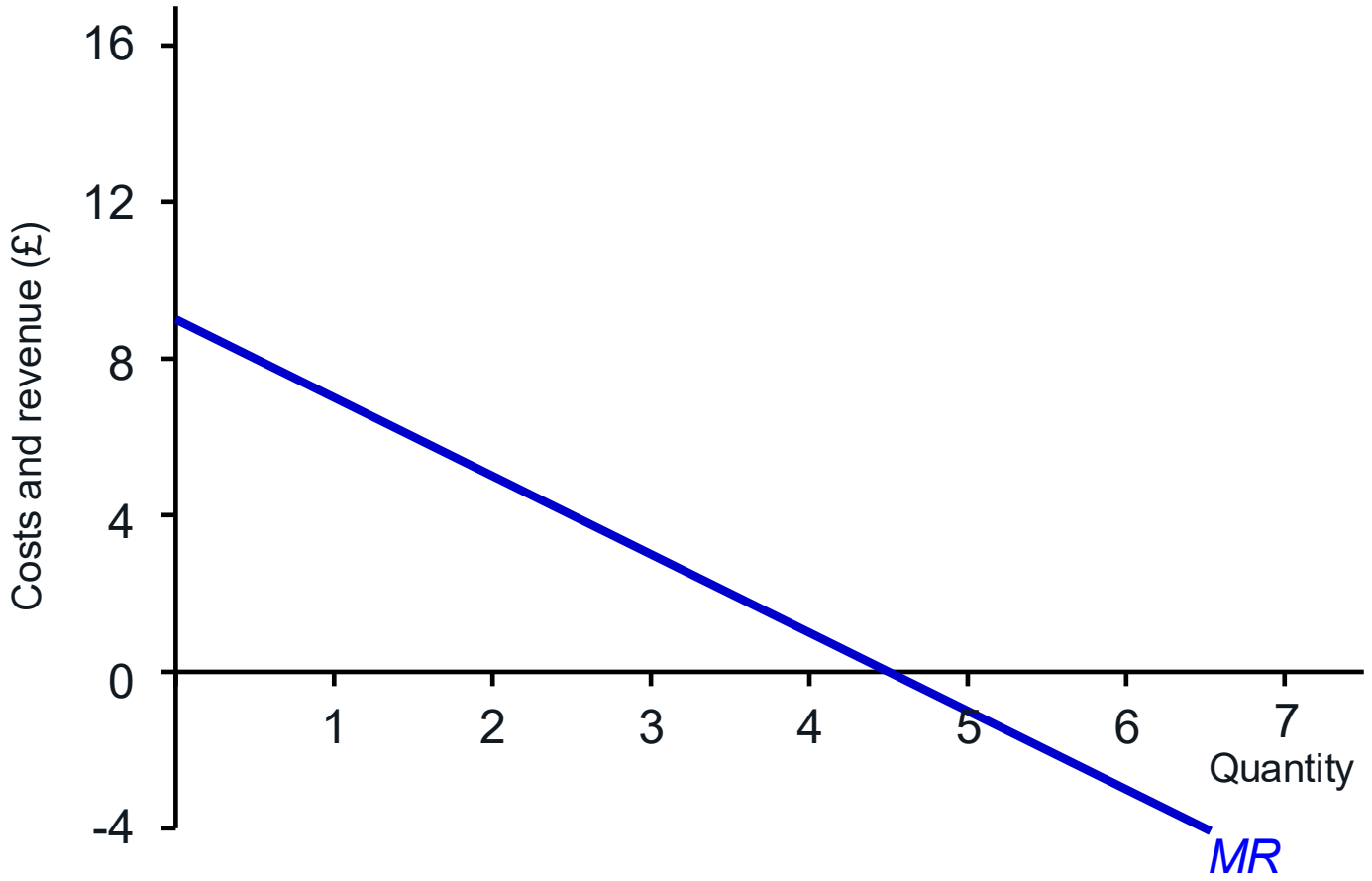
- **Stage 1**:
  profit maximised where **marginal revenue (*MR*) = marginal Cost (*MC*)**
  determine **the quantities of output that maximized profit Q_M**

- **Stage 2**:
  using Average Revenue *AR curve* *(determine price per unit of output)*
  and Average Cost *AC curve* *(determine cost per unit of output)*
  to measure maximum profit

$$\boldsymbol{Profit(T\Pi) = Total\ Revenue(TR) - Total\ Cost\ (TC)}$$

=Average Revenue (AR)*Q_M - Average Cost (AC) *Q_M

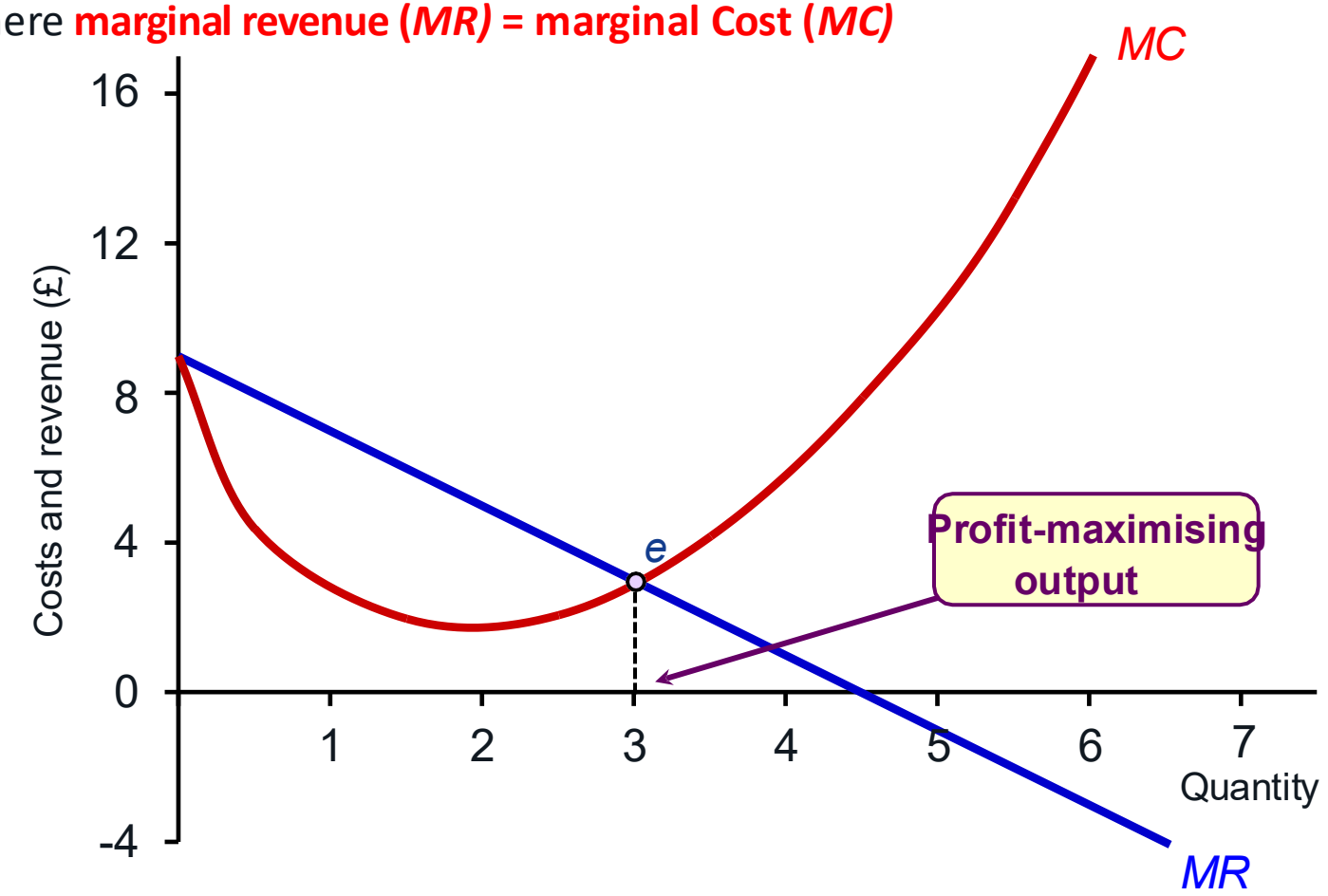
Profit-maximizing output using marginal curves



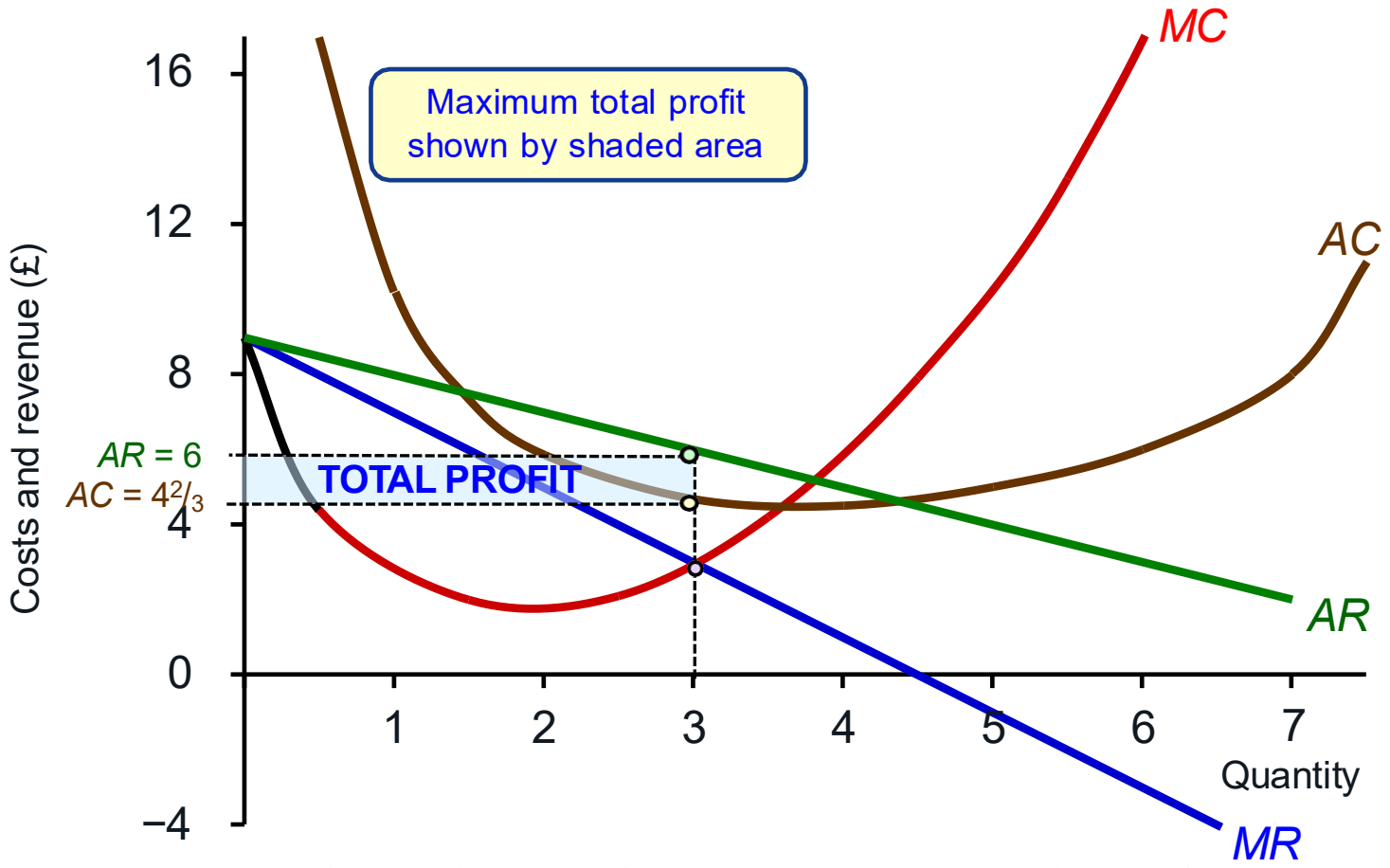
Profit-maximizing output using marginal curves

Stage 1:

profit maximised where **marginal revenue (*MR*) = marginal Cost (*MC*)**



Maximum profit using average curves



- **Stage 2**:

using *AR (determine price per unit of output)* and *AC (determine cost per unit of output)*curves to measure maximum profit

Reference

- Sloman, J., Wride, A. and Garratt,D (2018) “Economics” (10th edition)
- Gregory, Mankiw, N. (2017) “Economics” Cengage Textbooks